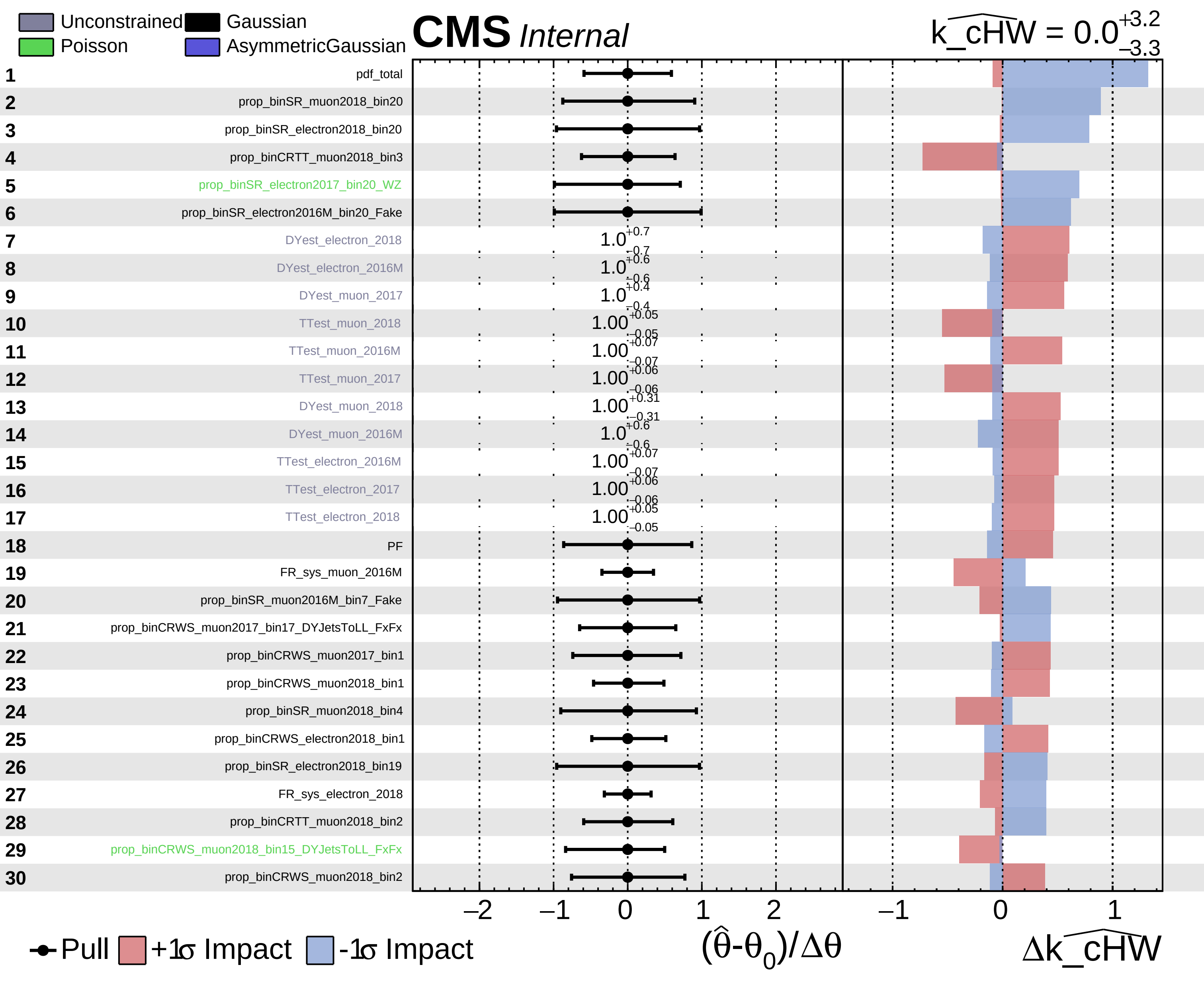
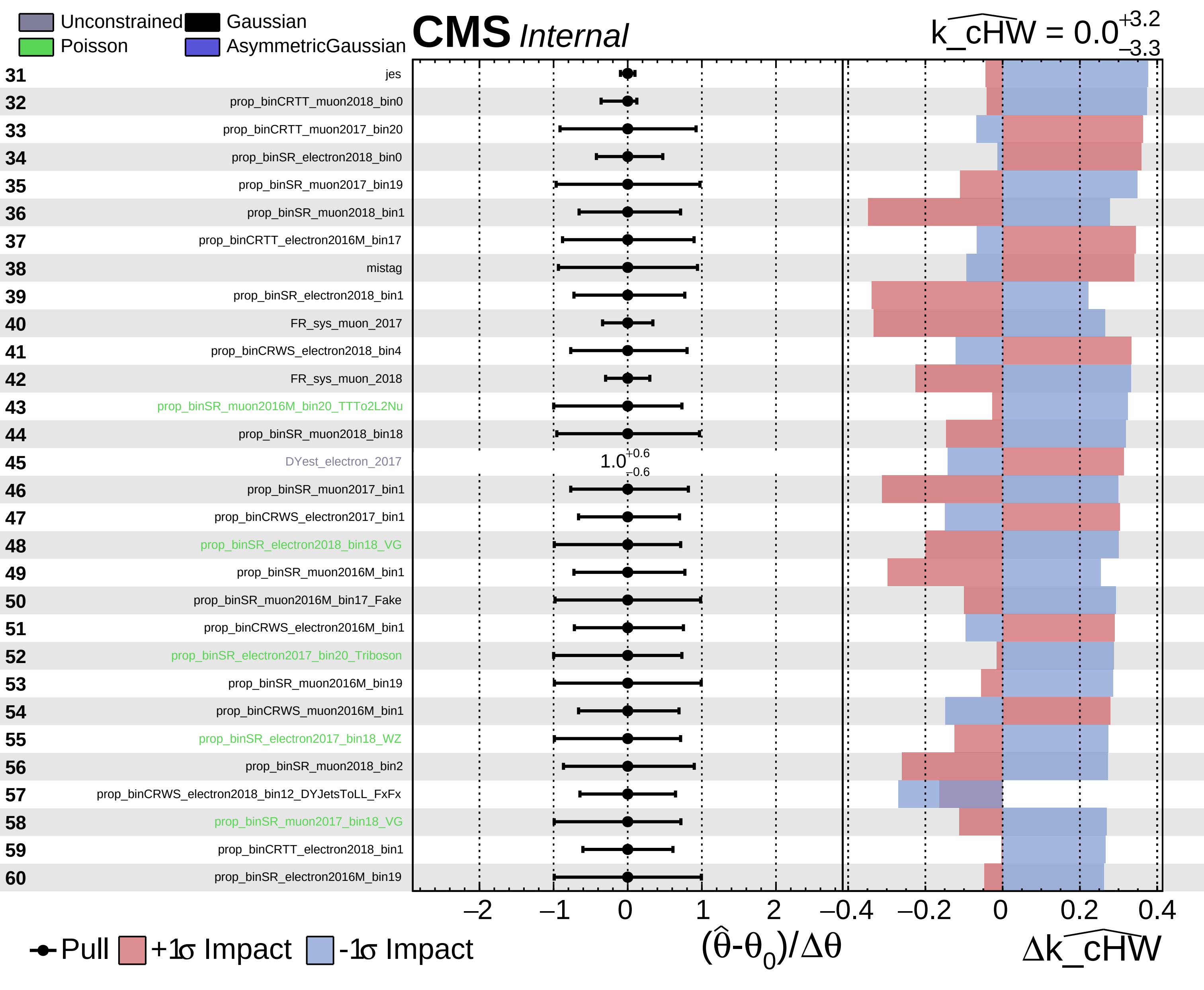
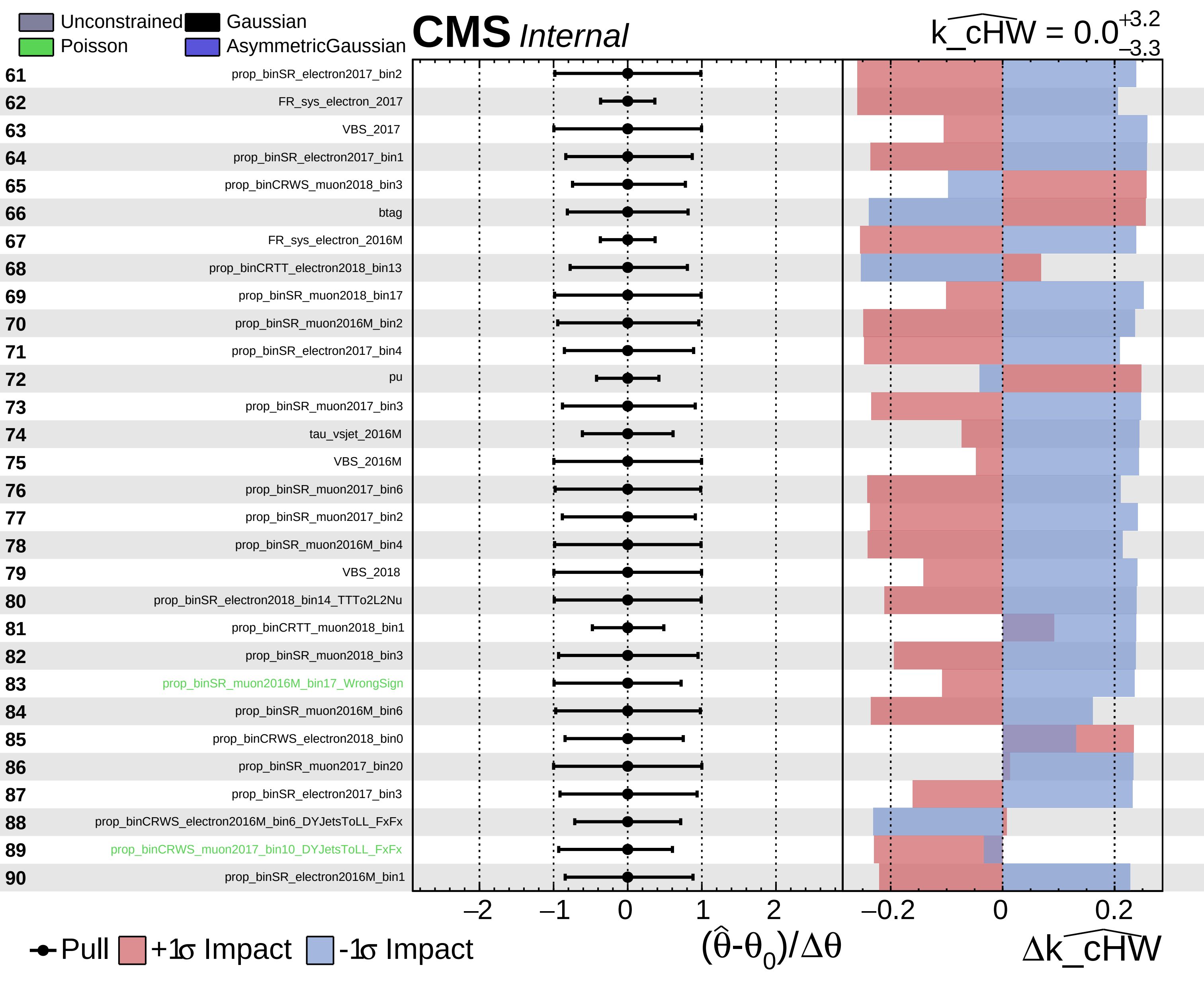


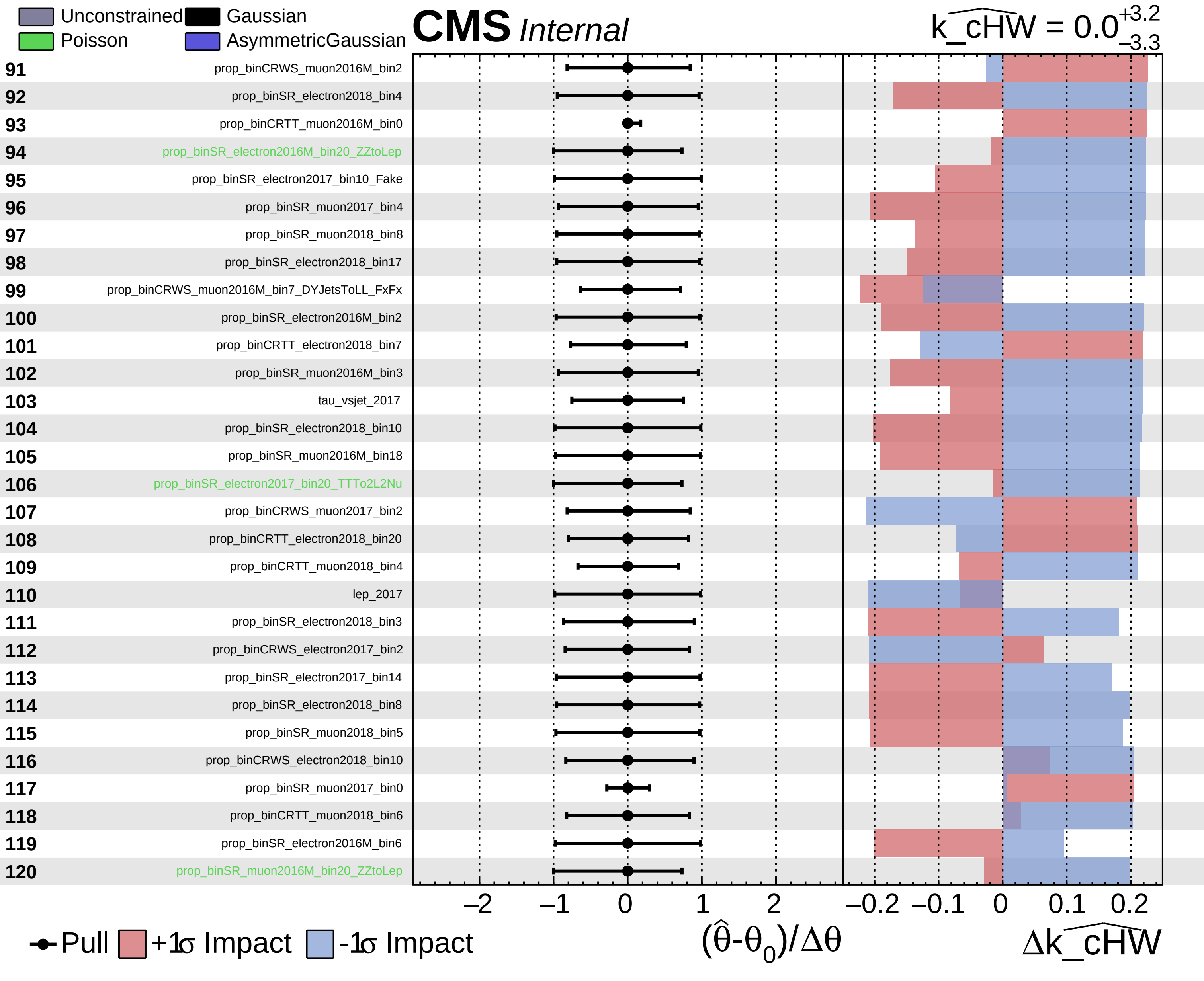
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
1 pdf_total
2 prop_binSR_muon2018_bin20
3 prop_binSR_electron2018_bin20
4 prop_binCRTT_muon2018_bin3
5 prop_binSR_electron2017_bin20_WZ
6 prop_binSR_electron2016M_bin20_Fake
7 DYest_electron_2018
8 DYest_electron_2016M
9 DYest_muon_2017
10 TTest_muon_2018
11 TTest_muon_2016M
12 TTest_muon_2017
13 DYest_muon_2018
14 DYest_muon_2016M
15 TTest_electron_2016M
16 TTest_electron_2017
17 TTest_electron_2018
18 PF
19 FR_sys_muon_2016M
20 prop_binSR_muon2016M_bin7_Fake
21 prop_binCRWS_muon2017_bin17_DYJetsToLL_FxFx
22 prop_binCRWS_muon2017_bin1
23 prop_binCRWS_muon2018_bin1
24 prop_binSR_muon2018_bin4
25 prop_binCRWS_electron2018_bin1
26 prop_binSR_electron2018_bin19
27 FR_sys_electron_2018
28 prop_binCRTT_muon2018_bin2
29 prop_binCRWS_muon2018_bin15_DYJetsToLL_FxFx
30 prop_binCRWS_muon2018_bin2
1.0 +0.7 −0.7
1.0 +0.6 −0.6
1.0 +0.4 −0.4
1.00 +0.05 −0.05
1.00 +0.07 −0.07
1.00 +0.06 −0.06
1.00 +0.31 −0.31
1.0 +0.6 −0.6
1.00 +0.07 −0.07
1.00 +0.06 −0.06
1.00 +0.05 −0.05
−2 −1 0 1 2
−1 0 1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW



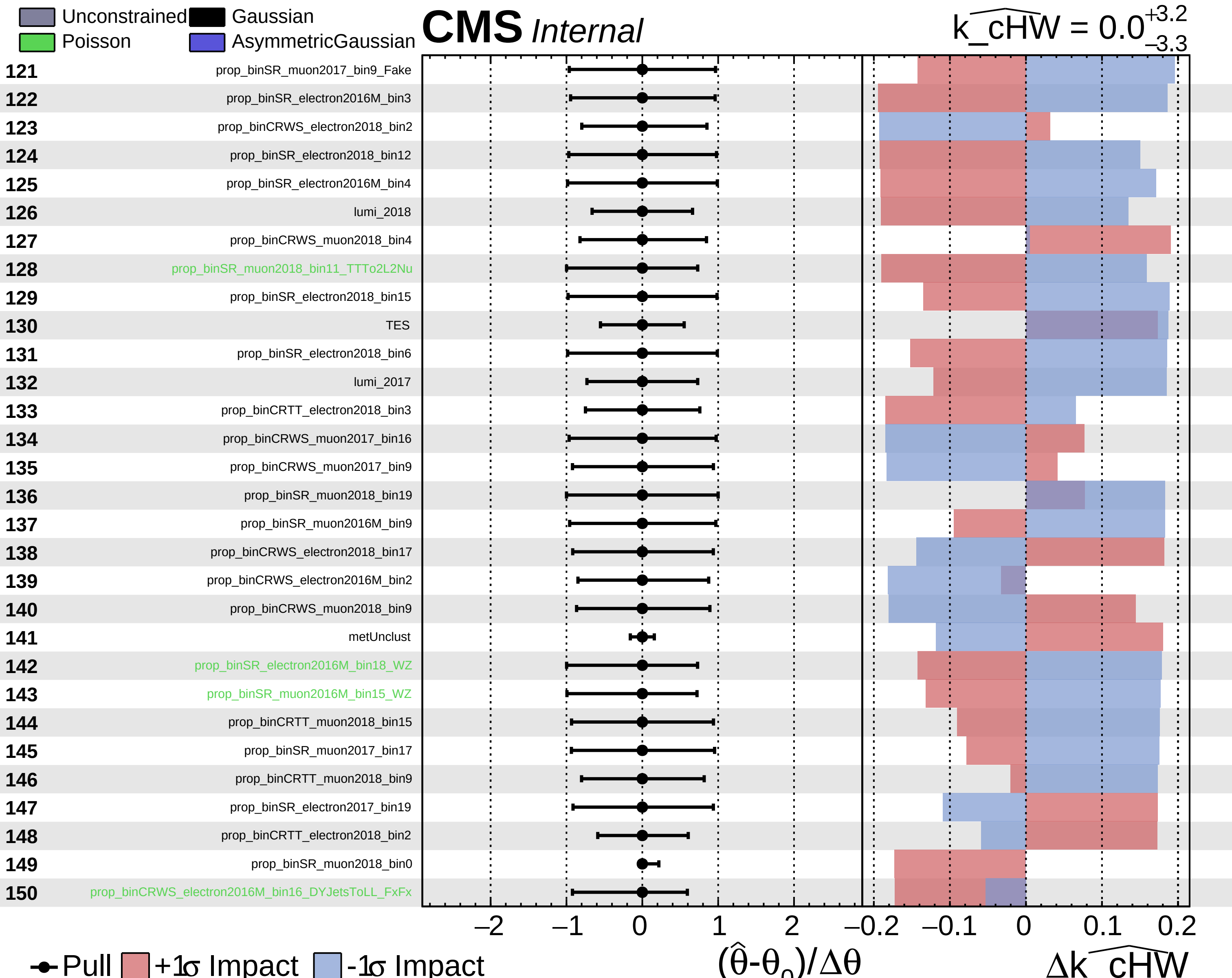
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0+3.2 −3.3
31 jes
32 prop_binCRTT_muon2018_bin0
33 prop_binCRTT_muon2017_bin20
34 prop_binSR_electron2018_bin0
35 prop_binSR_muon2017_bin19
36 prop_binSR_muon2018_bin1
37 prop_binCRTT_electron2016M_bin17
38 mistag
39 prop_binSR_electron2018_bin1
40 FR_sys_muon_2017
41 prop_binCRWS_electron2018_bin4
42 FR_sys_muon_2018
43 prop_binSR_muon2016M_bin20_TTTo2L2Nu
44 prop_binSR_muon2018_bin18
45 DYest_electron_2017
1.0+0.6 −0.6
46 prop_binSR_muon2017_bin1
47 prop_binCRWS_electron2017_bin1
48 prop_binSR_electron2018_bin18_VG
49 prop_binSR_muon2016M_bin1
50 prop_binSR_muon2016M_bin17_Fake
51 prop_binCRWS_electron2016M_bin1
52 prop_binSR_electron2017_bin20_Triboson
53 prop_binSR_muon2016M_bin19
54 prop_binCRWS_muon2016M_bin1
55 prop_binSR_electron2017_bin18_WZ
56 prop_binSR_muon2018_bin2
57 prop_binCRWS_electron2018_bin12_DYJetsToLL_FxFx
58 prop_binSR_muon2017_bin18_VG
59 prop_binCRTT_electron2018_bin1
60 prop_binSR_electron2016M_bin19
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

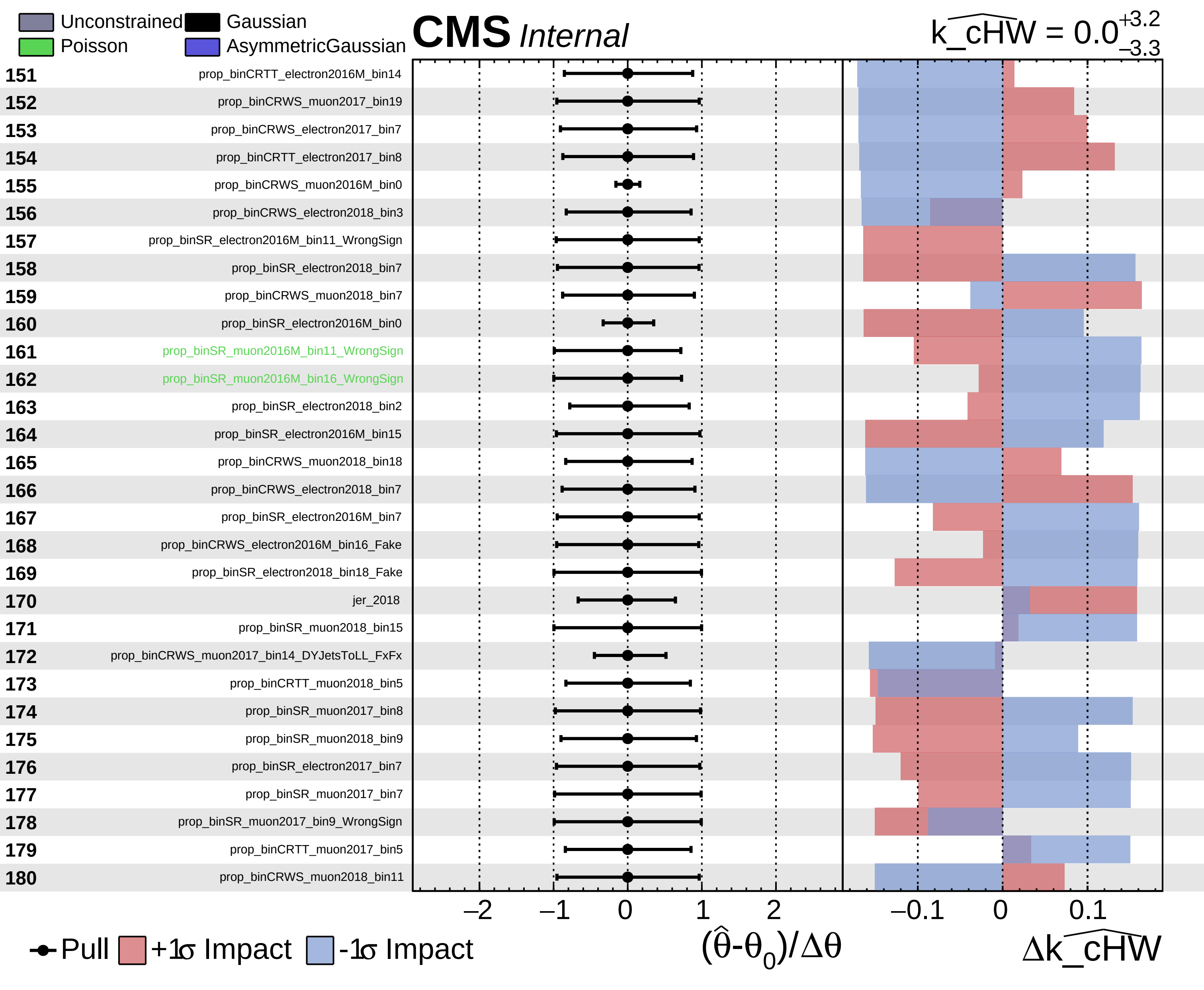


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
61 prop_binSR_electron2017_bin2
62 FR_sys_electron_2017
63 VBS_2017
64 prop_binSR_electron2017_bin1
65 prop_binCRWS_muon2018_bin3
66 btag
67 FR_sys_electron_2016M
68 prop_binCRTT_electron2018_bin13
69 prop_binSR_muon2018_bin17
70 prop_binSR_muon2016M_bin2
71 prop_binSR_electron2017_bin4
72 pu
73 prop_binSR_muon2017_bin3
74 tau_vsjet_2016M
75 VBS_2016M
76 prop_binSR_muon2017_bin6
77 prop_binSR_muon2017_bin2
78 prop_binSR_muon2016M_bin4
79 VBS_2018
80 prop_binSR_electron2018_bin14_TTTo2L2Nu
81 prop_binCRTT_muon2018_bin1
82 prop_binSR_muon2018_bin3
83 prop_binSR_muon2016M_bin17_WrongSign
84 prop_binSR_muon2016M_bin6
85 prop_binCRWS_electron2018_bin0
86 prop_binSR_muon2017_bin20
87 prop_binSR_electron2017_bin3
88 prop_binCRWS_electron2016M_bin6_DYJetsToLL_FxFx
89 prop_binCRWS_muon2017_bin10_DYJetsToLL_FxFx
90 prop_binSR_electron2016M_bin1
−2
−1
0
1
2
−0.2
0
0.2
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

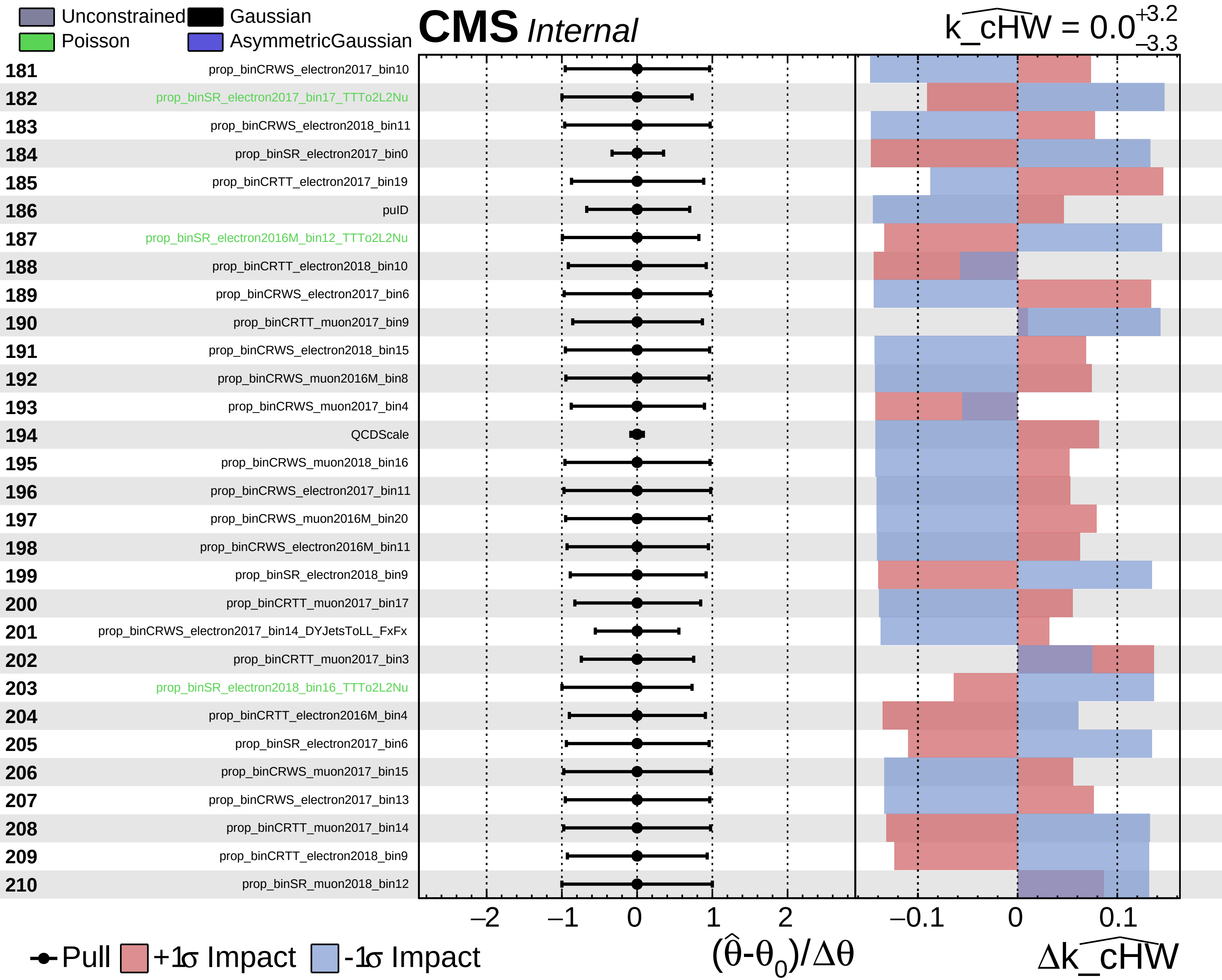


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+3.2}_{-3.3}$
91 prop_binCRWS_muon2016M_bin2
92 prop_binSR_electron2018_bin4
93 prop_binCRTT_muon2016M_bin0
94 prop_binSR_electron2016M_bin20_ZZtoLep
95 prop_binSR_electron2017_bin10_Fake
96 prop_binSR_muon2017_bin4
97 prop_binSR_muon2018_bin8
98 prop_binSR_electron2018_bin17
99 prop_binCRWS_muon2016M_bin7_DYJetsToLL_FxFx
100 prop_binSR_electron2016M_bin2
101 prop_binCRTT_electron2018_bin7
102 prop_binSR_muon2016M_bin3
103 tau_vsjet_2017
104 prop_binSR_electron2018_bin10
105 prop_binSR_muon2016M_bin18
106 prop_binSR_electron2017_bin20_TTTo2L2Nu
107 prop_binCRWS_muon2017_bin2
108 prop_binCRTT_electron2018_bin20
109 prop_binCRTT_muon2018_bin4
110 lep_2017
111 prop_binSR_electron2018_bin3
112 prop_binCRWS_electron2017_bin2
113 prop_binSR_electron2017_bin14
114 prop_binSR_electron2018_bin8
115 prop_binSR_muon2018_bin5
116 prop_binCRWS_electron2018_bin10
117 prop_binSR_muon2017_bin0
118 prop_binCRTT_muon2018_bin6
119 prop_binSR_electron2016M_bin6
120 prop_binSR_muon2016M_bin20_ZZtoLep
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

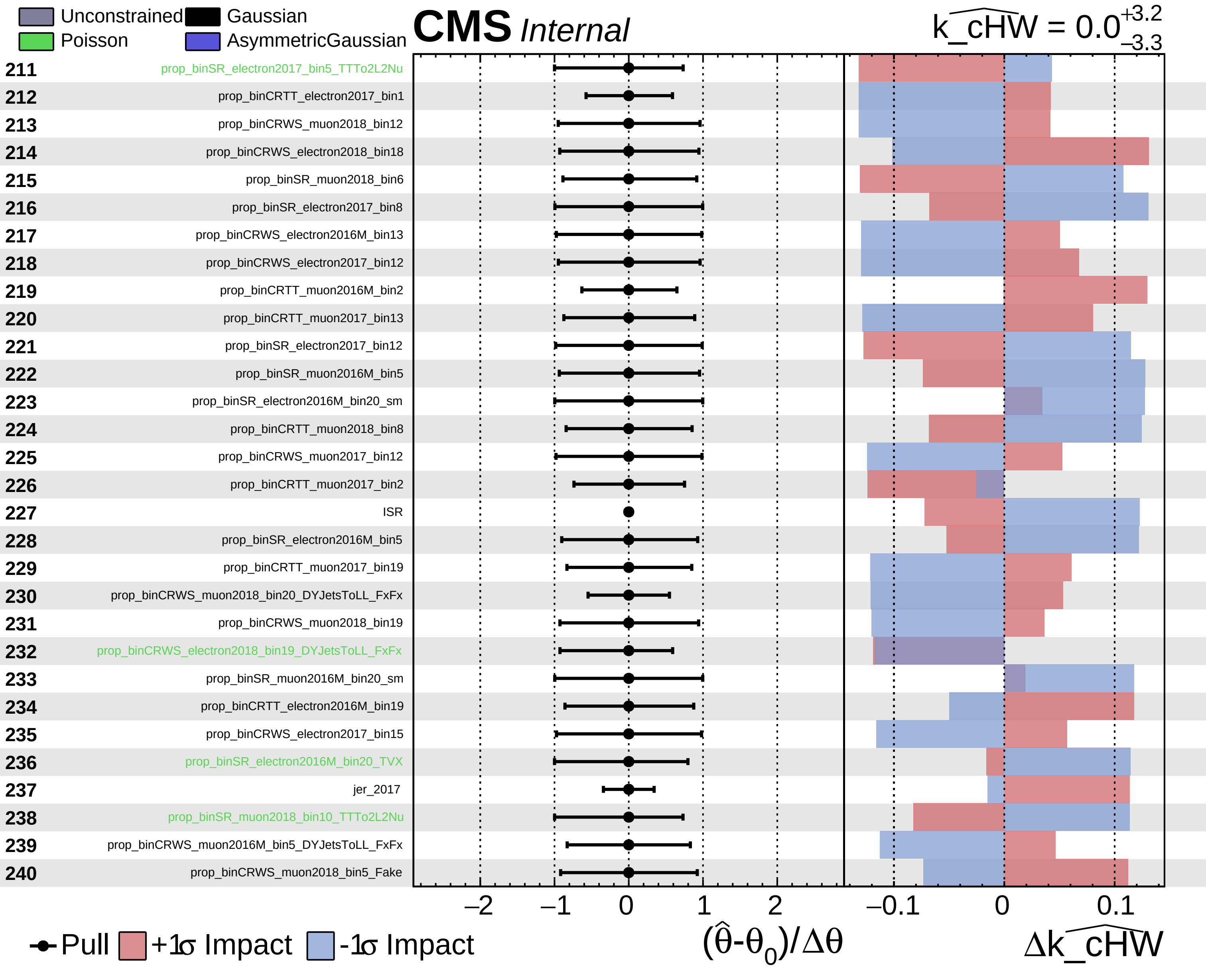

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
121 prop_binSR_muon2017_bin9_Fake
122 prop_binSR_electron2016M_bin3
123 prop_binCRWS_electron2018_bin2
124 prop_binSR_electron2018_bin12
125 prop_binSR_electron2016M_bin4
126 lumi_2018
127 prop_binCRWS_muon2018_bin4
128 prop_binSR_muon2018_bin11_TTTo2L2Nu
129 prop_binSR_electron2018_bin15
130 TES
131 prop_binSR_electron2018_bin6
132 lumi_2017
133 prop_binCRTT_electron2018_bin3
134 prop_binCRWS_muon2017_bin16
135 prop_binCRWS_muon2017_bin9
136 prop_binSR_muon2018_bin19
137 prop_binSR_muon2016M_bin9
138 prop_binCRWS_electron2018_bin17
139 prop_binCRWS_electron2016M_bin2
140 prop_binCRWS_muon2018_bin9
141 metUnclust
142 prop_binSR_electron2016M_bin18_WZ
143 prop_binSR_muon2016M_bin15_WZ
144 prop_binCRTT_muon2018_bin15
145 prop_binSR_muon2017_bin17
146 prop_binCRTT_muon2018_bin9
147 prop_binSR_electron2017_bin19
148 prop_binCRTT_electron2018_bin2
149 prop_binSR_muon2018_bin0
150 prop_binCRWS_electron2016M_bin16_DYJetsToLL_FxFx
−2
−1
0
1
2
−0.2
−0.1
0
0.1
0.2
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



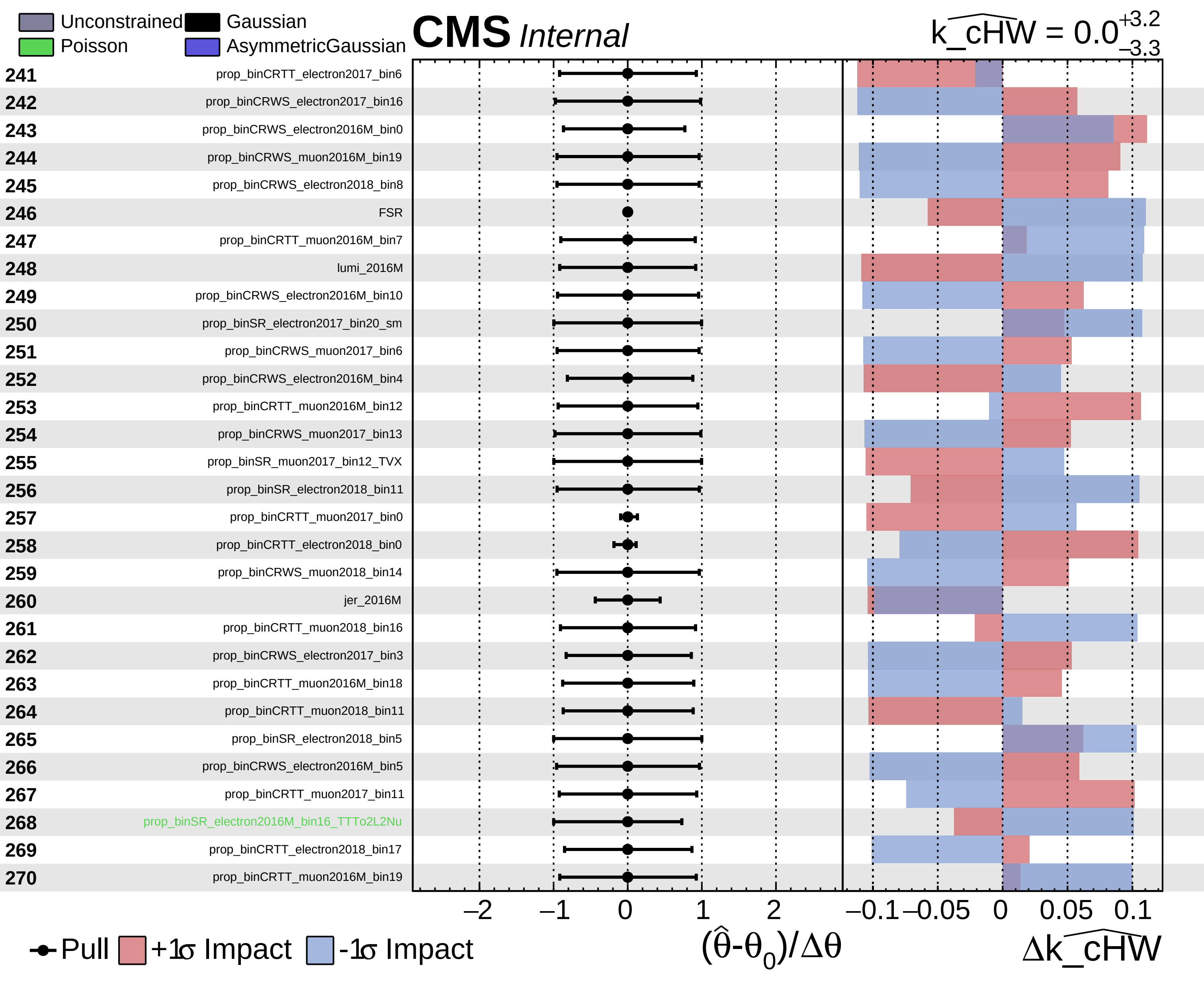
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
151 prop_binCRTT_electron2016M_bin14
152 prop_binCRWS_muon2017_bin19
153 prop_binCRWS_electron2017_bin7
154 prop_binCRTT_electron2017_bin8
155 prop_binCRWS_muon2016M_bin0
156 prop_binCRWS_electron2018_bin3
157 prop_binSR_electron2016M_bin11_WrongSign
158 prop_binSR_electron2018_bin7
159 prop_binCRWS_muon2018_bin7
160 prop_binSR_electron2016M_bin0
161 prop_binSR_muon2016M_bin11_WrongSign
162 prop_binSR_muon2016M_bin16_WrongSign
163 prop_binSR_electron2018_bin2
164 prop_binSR_electron2016M_bin15
165 prop_binCRWS_muon2018_bin18
166 prop_binCRWS_electron2018_bin7
167 prop_binSR_electron2016M_bin7
168 prop_binCRWS_electron2016M_bin16_Fake
169 prop_binSR_electron2018_bin18_Fake
170 jer_2018
171 prop_binSR_muon2018_bin15
172 prop_binCRWS_muon2017_bin14_DYJetsToLL_FxFx
173 prop_binCRTT_muon2018_bin5
174 prop_binSR_muon2017_bin8
175 prop_binSR_muon2018_bin9
176 prop_binSR_electron2017_bin7
177 prop_binSR_muon2017_bin7
178 prop_binSR_muon2017_bin9_WrongSign
179 prop_binCRTT_muon2017_bin5
180 prop_binCRWS_muon2018_bin11
−2
−1
0
1
2
−0.1
0
0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW

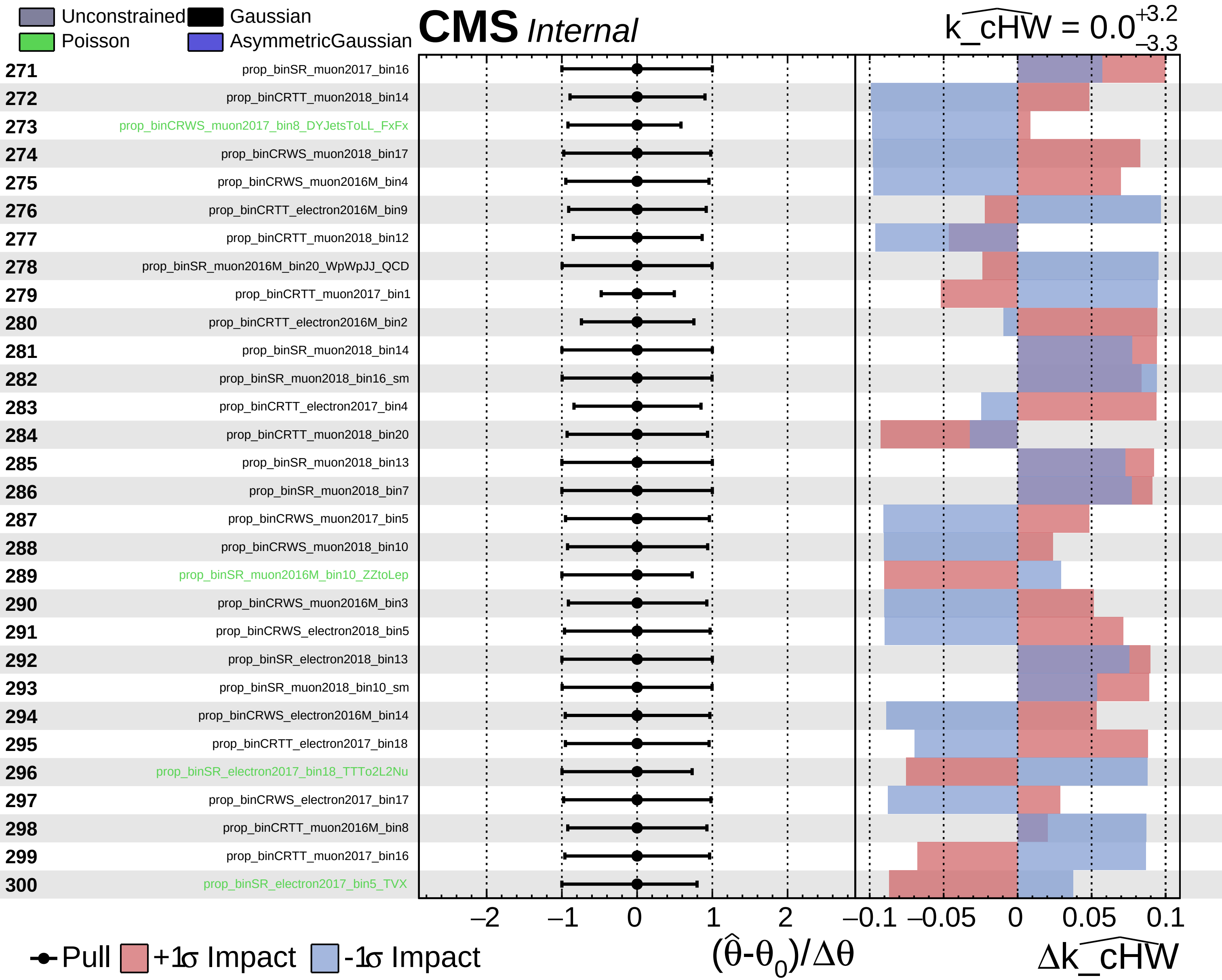


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
181 prop_binCRWS_electron2017_bin10
182 prop_binSR_electron2017_bin17_TTTo2L2Nu
183 prop_binCRWS_electron2018_bin11
184 prop_binSR_electron2017_bin0
185 prop_binCRTT_electron2017_bin19
186 puID
187 prop_binSR_electron2016M_bin12_TTTo2L2Nu
188 prop_binCRTT_electron2018_bin10
189 prop_binCRWS_electron2017_bin6
190 prop_binCRTT_muon2017_bin9
191 prop_binCRWS_electron2018_bin15
192 prop_binCRWS_muon2016M_bin8
193 prop_binCRWS_muon2017_bin4
194 QCDScale
195 prop_binCRWS_muon2018_bin16
196 prop_binCRWS_electron2017_bin11
197 prop_binCRWS_muon2016M_bin20
198 prop_binCRWS_electron2016M_bin11
199 prop_binSR_electron2018_bin9
200 prop_binCRTT_muon2017_bin17
201 prop_binCRWS_electron2017_bin14_DYJetsToLL_FxFx
202 prop_binCRTT_muon2017_bin3
203 prop_binSR_electron2018_bin16_TTTo2L2Nu
204 prop_binCRTT_electron2016M_bin4
205 prop_binSR_electron2017_bin6
206 prop_binCRWS_muon2017_bin15
207 prop_binCRWS_electron2017_bin13
208 prop_binCRTT_muon2017_bin14
209 prop_binCRTT_electron2018_bin9
210 prop_binSR_muon2018_bin12
−2
−1
0
1
2
−0.1
0
0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW

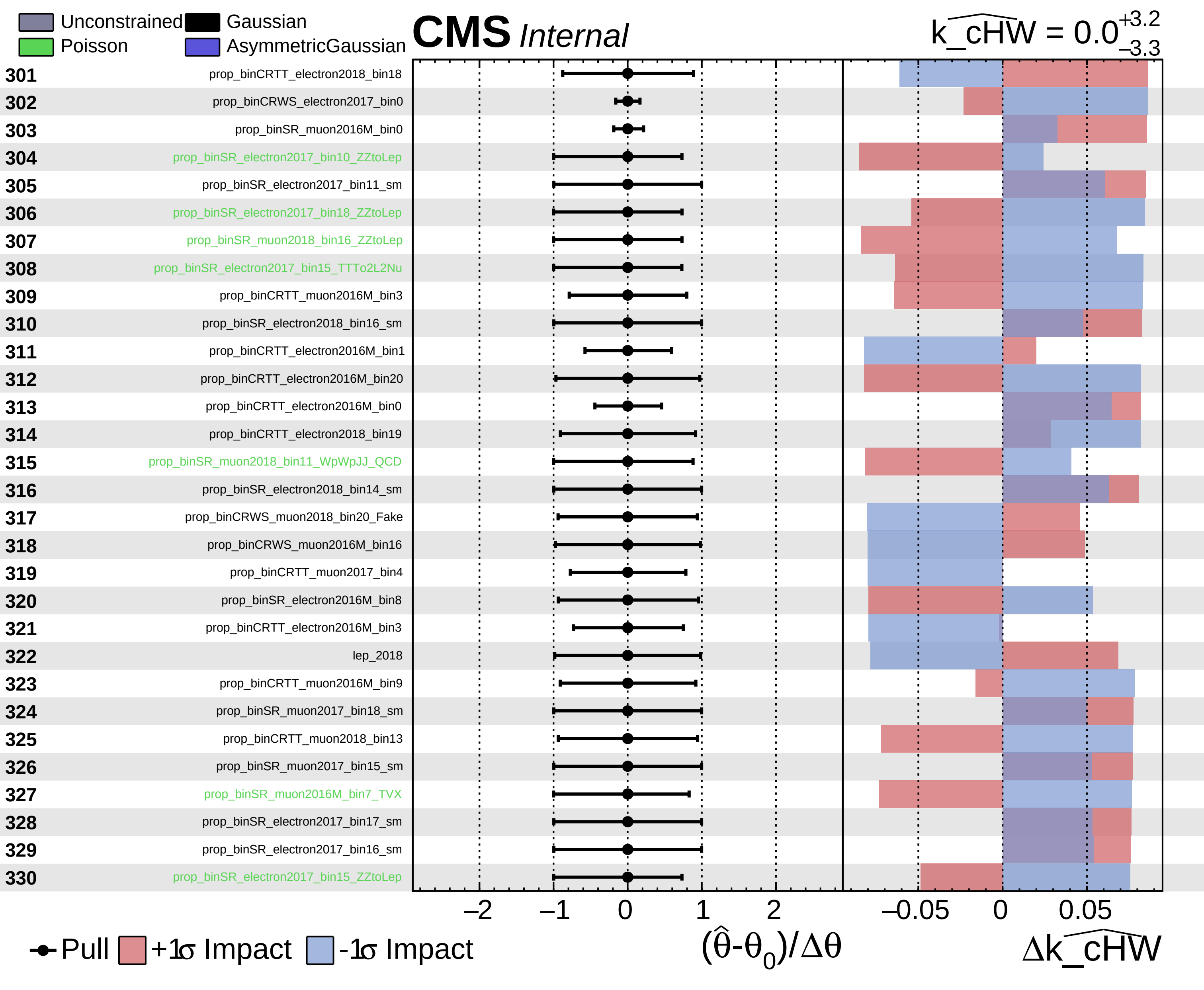


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
211 prop_binSR_electron2017_bin5_TTTo2L2Nu
212 prop_binCRTT_electron2017_bin1
213 prop_binCRWS_muon2018_bin12
214 prop_binCRWS_electron2018_bin18
215 prop_binSR_muon2018_bin6
216 prop_binSR_electron2017_bin8
217 prop_binCRWS_electron2016M_bin13
218 prop_binCRWS_electron2017_bin12
219 prop_binCRTT_muon2016M_bin2
220 prop_binCRTT_muon2017_bin13
221 prop_binSR_electron2017_bin12
222 prop_binSR_muon2016M_bin5
223 prop_binSR_electron2016M_bin20_sm
224 prop_binCRTT_muon2018_bin8
225 prop_binCRWS_muon2017_bin12
226 prop_binCRTT_muon2017_bin2
227 ISR
228 prop_binSR_electron2016M_bin5
229 prop_binCRTT_muon2017_bin19
230 prop_binCRWS_muon2018_bin20_DYJetsToLL_FxFx
231 prop_binCRWS_muon2018_bin19
232 prop_binCRWS_electron2018_bin19_DYJetsToLL_FxFx
233 prop_binSR_muon2016M_bin20_sm
234 prop_binCRTT_electron2016M_bin19
235 prop_binCRWS_electron2017_bin15
236 prop_binSR_electron2016M_bin20_TVX
237 jer_2017
238 prop_binSR_muon2018_bin10_TTTo2L2Nu
239 prop_binCRWS_muon2016M_bin5_DYJetsToLL_FxFx
240 prop_binCRWS_muon2018_bin5_Fake
−2 −1 0 1 2
−0.1 0 0.1
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

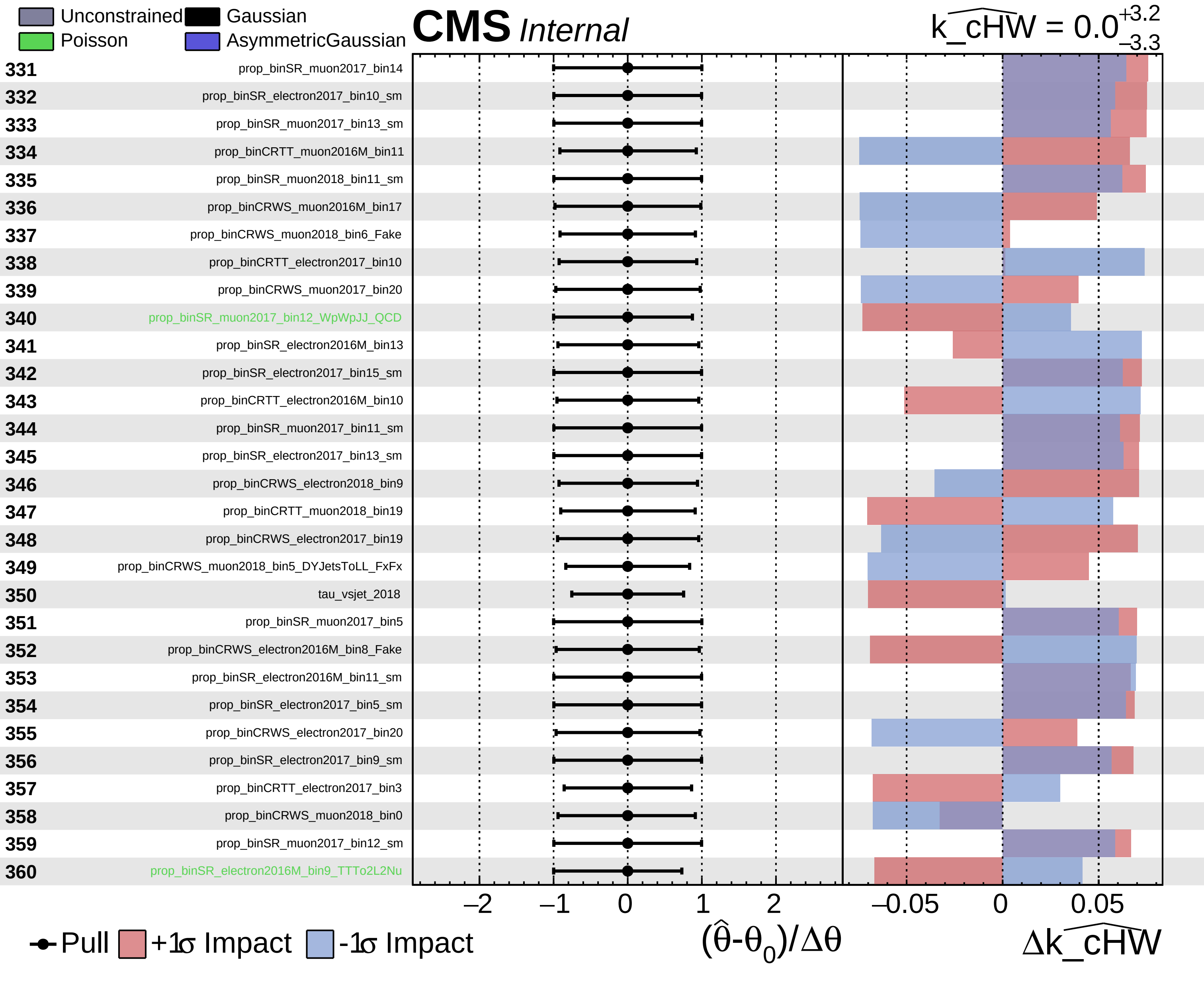

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+3.2}_{-3.3}$
241 prop_binCRTT_electron2017_bin6
242 prop_binCRWS_electron2017_bin16
243 prop_binCRWS_electron2016M_bin0
244 prop_binCRWS_muon2016M_bin19
245 prop_binCRWS_electron2018_bin8
246 FSR
247 prop_binCRTT_muon2016M_bin7
248 lumi_2016M
249 prop_binCRWS_electron2016M_bin10
250 prop_binSR_electron2017_bin20_sm
251 prop_binCRWS_muon2017_bin6
252 prop_binCRWS_electron2016M_bin4
253 prop_binCRTT_muon2016M_bin12
254 prop_binCRWS_muon2017_bin13
255 prop_binSR_muon2017_bin12_TVX
256 prop_binSR_electron2018_bin11
257 prop_binCRTT_muon2017_bin0
258 prop_binCRTT_electron2018_bin0
259 prop_binCRWS_muon2018_bin14
260 jer_2016M
261 prop_binCRTT_muon2018_bin16
262 prop_binCRWS_electron2017_bin3
263 prop_binCRTT_muon2016M_bin18
264 prop_binCRTT_muon2018_bin11
265 prop_binSR_electron2018_bin5
266 prop_binCRWS_electron2016M_bin5
267 prop_binCRTT_muon2017_bin11
268 prop_binSR_electron2016M_bin16_TTTo2L2Nu
269 prop_binCRTT_electron2018_bin17
270 prop_binCRTT_muon2016M_bin19
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



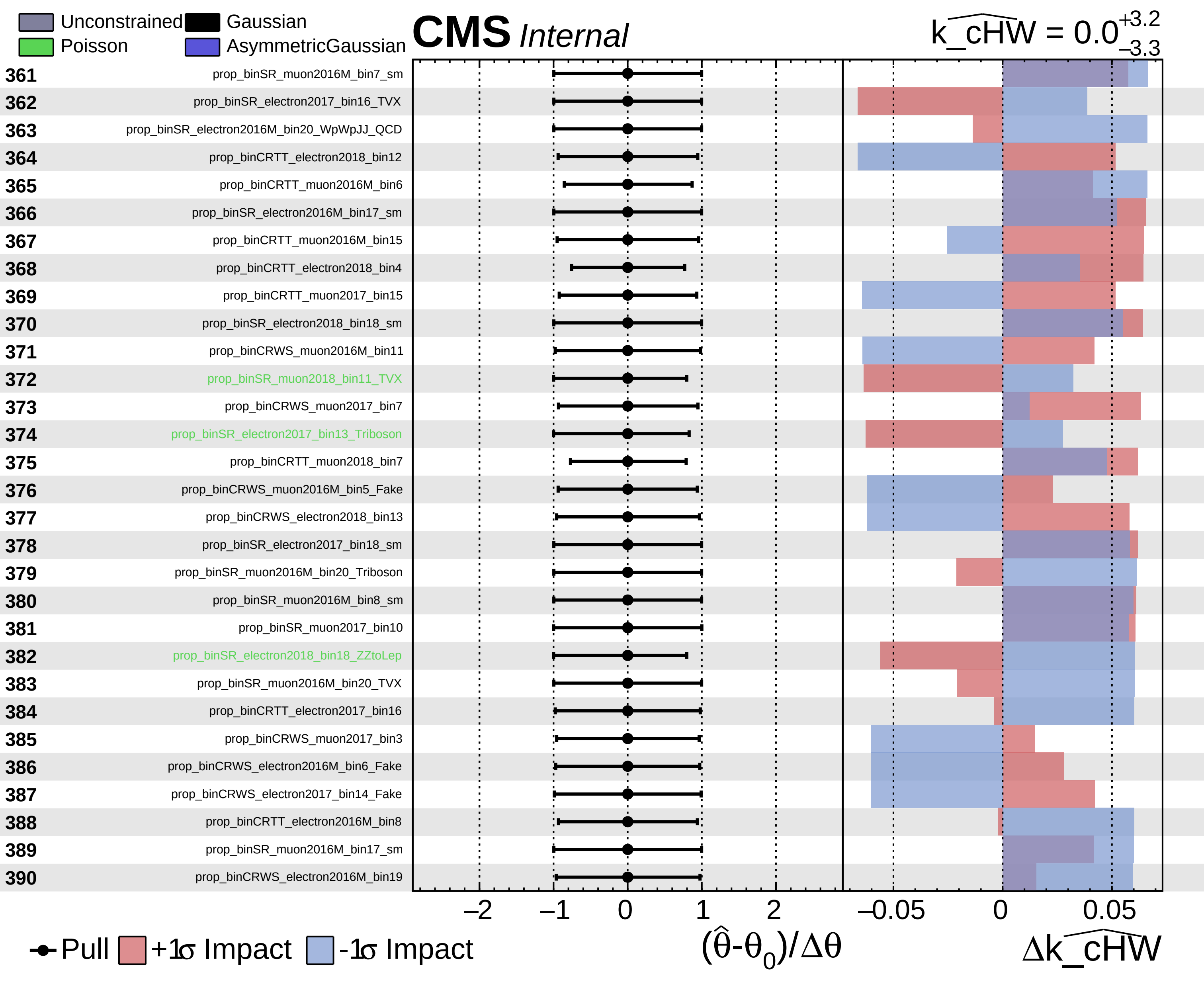
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+3.2}_{-3.3}$
271 prop_binSR_muon2017_bin16
272 prop_binCRTT_muon2018_bin14
273 prop_binCRWS_muon2017_bin8_DYJetsToLL_FxFx
274 prop_binCRWS_muon2018_bin17
275 prop_binCRWS_muon2016M_bin4
276 prop_binCRTT_electron2016M_bin9
277 prop_binCRTT_muon2018_bin12
278 prop_binSR_muon2016M_bin20_WpWpJJ_QCD
279 prop_binCRTT_muon2017_bin1
280 prop_binCRTT_electron2016M_bin2
281 prop_binSR_muon2018_bin14
282 prop_binSR_muon2018_bin16_sm
283 prop_binCRTT_electron2017_bin4
284 prop_binCRTT_muon2018_bin20
285 prop_binSR_muon2018_bin13
286 prop_binSR_muon2018_bin7
287 prop_binCRWS_muon2017_bin5
288 prop_binCRWS_muon2018_bin10
289 prop_binSR_muon2016M_bin10_ZZtoLep
290 prop_binCRWS_muon2016M_bin3
291 prop_binCRWS_electron2018_bin5
292 prop_binSR_electron2018_bin13
293 prop_binSR_muon2018_bin10_sm
294 prop_binCRWS_electron2016M_bin14
295 prop_binCRTT_electron2017_bin18
296 prop_binSR_electron2017_bin18_TTTo2L2Nu
297 prop_binCRWS_electron2017_bin17
298 prop_binCRTT_muon2016M_bin8
299 prop_binCRTT_muon2017_bin16
300 prop_binSR_electron2017_bin5_TVX
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



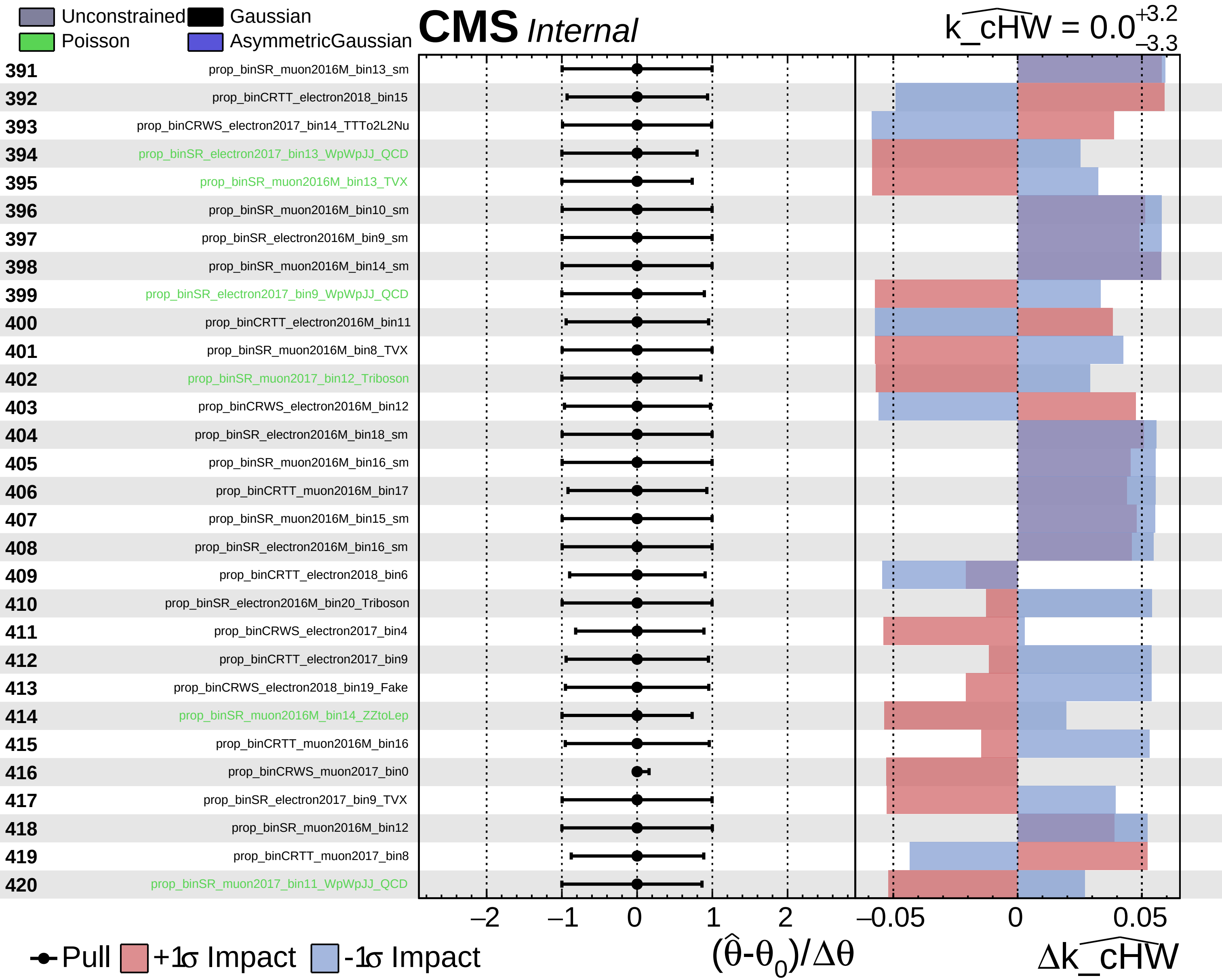
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+3.2}_{-3.3}$
301 prop_binCRTT_electron2018_bin18
302 prop_binCRWS_electron2017_bin0
303 prop_binSR_muon2016M_bin0
304 prop_binSR_electron2017_bin10_ZZtoLep
305 prop_binSR_electron2017_bin11_sm
306 prop_binSR_electron2017_bin18_ZZtoLep
307 prop_binSR_muon2018_bin16_ZZtoLep
308 prop_binSR_electron2017_bin15_TTTo2L2Nu
309 prop_binCRTT_muon2016M_bin3
310 prop_binSR_electron2018_bin16_sm
311 prop_binCRTT_electron2016M_bin1
312 prop_binCRTT_electron2016M_bin20
313 prop_binCRTT_electron2016M_bin0
314 prop_binCRTT_electron2018_bin19
315 prop_binSR_muon2018_bin11_WpWpJJ_QCD
316 prop_binSR_electron2018_bin14_sm
317 prop_binCRWS_muon2018_bin20_Fake
318 prop_binCRWS_muon2016M_bin16
319 prop_binCRTT_muon2017_bin4
320 prop_binSR_electron2016M_bin8
321 prop_binCRTT_electron2016M_bin3
322 lep_2018
323 prop_binCRTT_muon2016M_bin9
324 prop_binSR_muon2017_bin18_sm
325 prop_binCRTT_muon2018_bin13
326 prop_binSR_muon2017_bin15_sm
327 prop_binSR_muon2016M_bin7_TVX
328 prop_binSR_electron2017_bin17_sm
329 prop_binSR_electron2017_bin16_sm
330 prop_binSR_electron2017_bin15_ZZtoLep
−2
−1
0
1
2
−0.05
0
0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



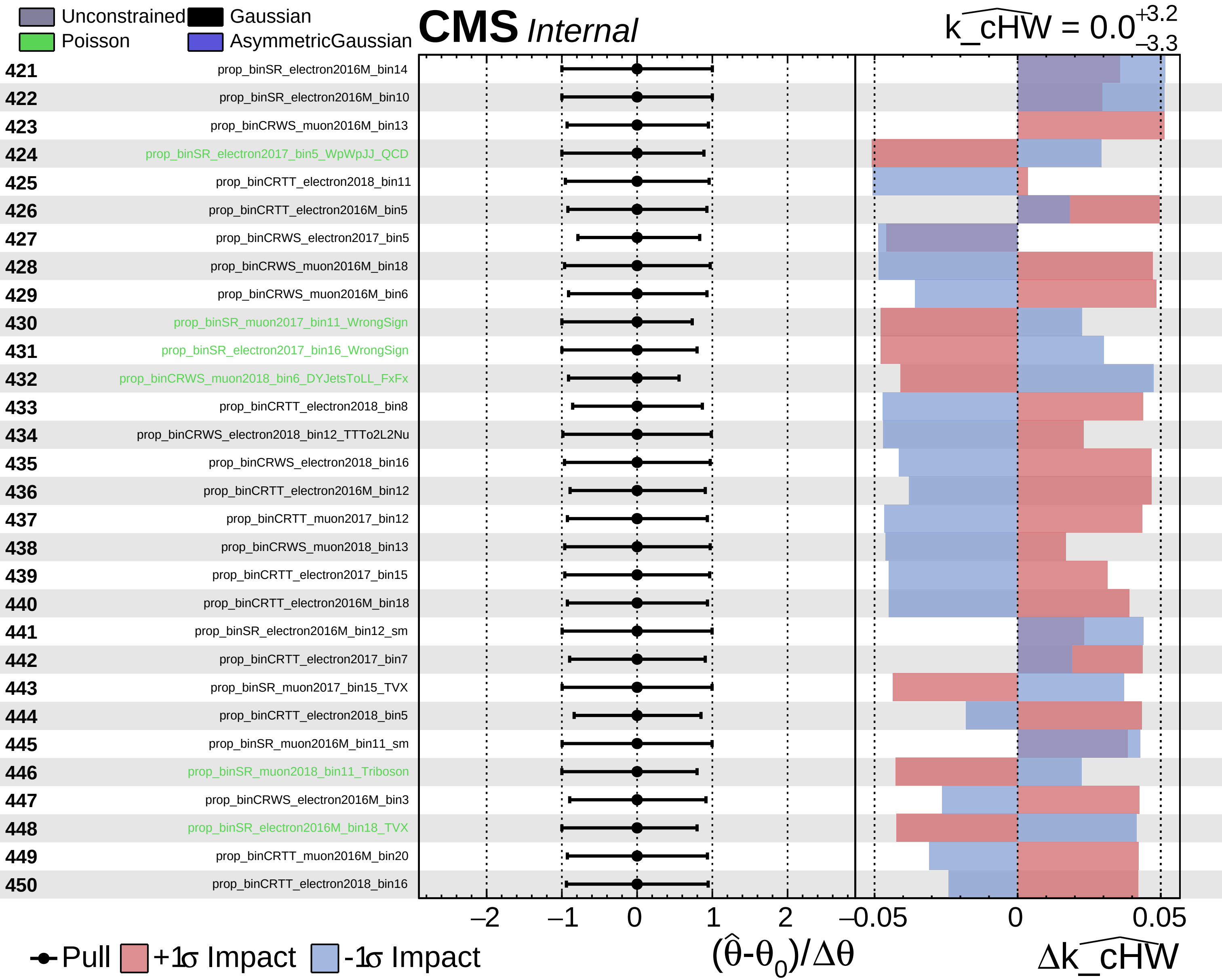
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+3.2}_{-3.3}$
331 prop_binSR_muon2017_bin14
332 prop_binSR_electron2017_bin10_sm
333 prop_binSR_muon2017_bin13_sm
334 prop_binCRTT_muon2016M_bin11
335 prop_binSR_muon2018_bin11_sm
336 prop_binCRWS_muon2016M_bin17
337 prop_binCRWS_muon2018_bin6_Fake
338 prop_binCRTT_electron2017_bin10
339 prop_binCRWS_muon2017_bin20
340 prop_binSR_muon2017_bin12_WpWpJJ_QCD
341 prop_binSR_electron2016M_bin13
342 prop_binSR_electron2017_bin15_sm
343 prop_binCRTT_electron2016M_bin10
344 prop_binSR_muon2017_bin11_sm
345 prop_binSR_electron2017_bin13_sm
346 prop_binCRWS_electron2018_bin9
347 prop_binCRTT_muon2018_bin19
348 prop_binCRWS_electron2017_bin19
349 prop_binCRWS_muon2018_bin5_DYJetsToLL_FxFx
350 tau_vsjet_2018
351 prop_binSR_muon2017_bin5
352 prop_binCRWS_electron2016M_bin8_Fake
353 prop_binSR_electron2016M_bin11_sm
354 prop_binSR_electron2017_bin5_sm
355 prop_binCRWS_electron2017_bin20
356 prop_binSR_electron2017_bin9_sm
357 prop_binCRTT_electron2017_bin3
358 prop_binCRWS_muon2018_bin0
359 prop_binSR_muon2017_bin12_sm
360 prop_binSR_electron2016M_bin9_TTTo2L2Nu
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



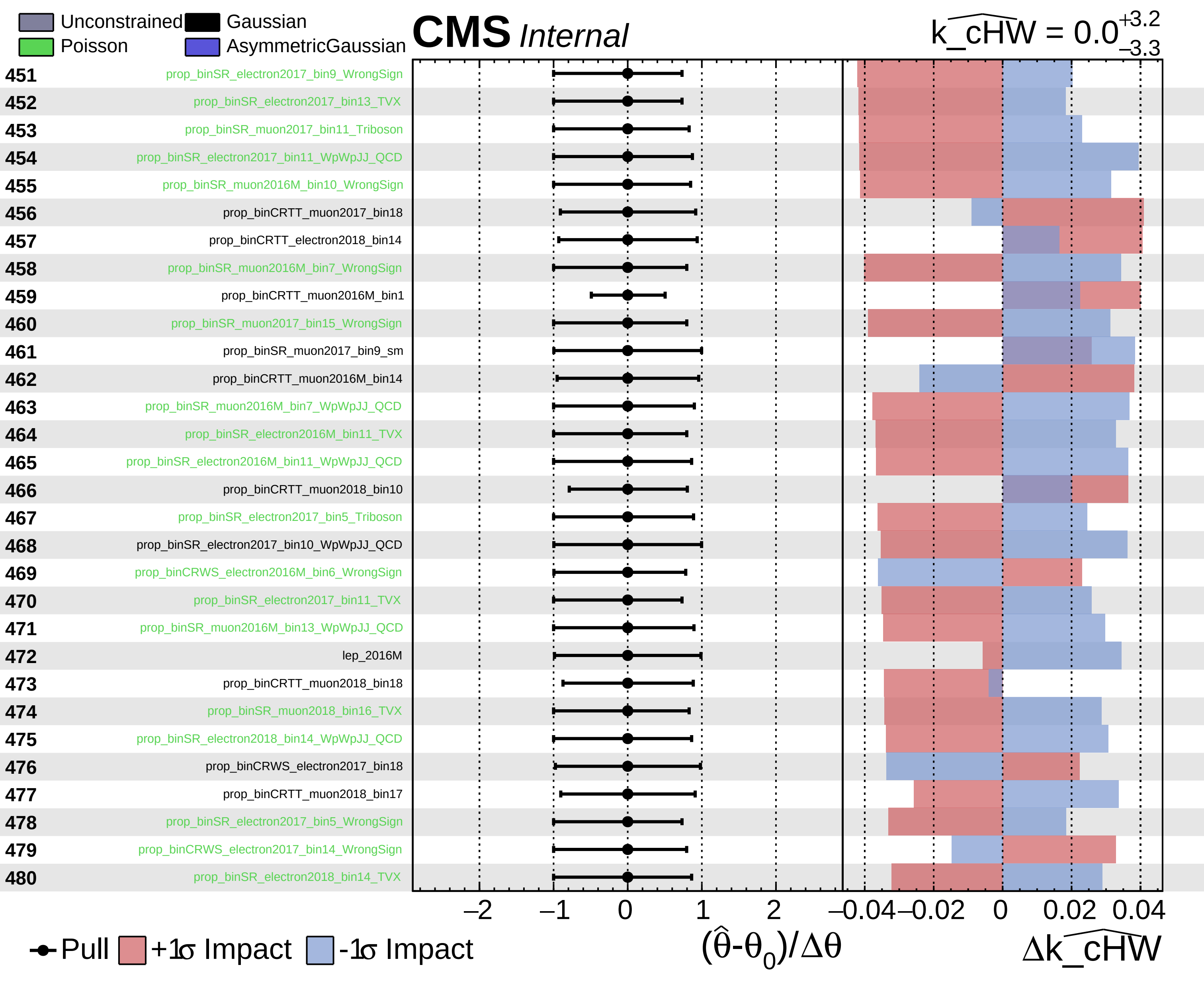
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
361 prop_binSR_muon2016M_bin7_sm
362 prop_binSR_electron2017_bin16_TVX
363 prop_binSR_electron2016M_bin20_WpWpJJ_QCD
364 prop_binCRTT_electron2018_bin12
365 prop_binCRTT_muon2016M_bin6
366 prop_binSR_electron2016M_bin17_sm
367 prop_binCRTT_muon2016M_bin15
368 prop_binCRTT_electron2018_bin4
369 prop_binCRTT_muon2017_bin15
370 prop_binSR_electron2018_bin18_sm
371 prop_binCRWS_muon2016M_bin11
372 prop_binSR_muon2018_bin11_TVX
373 prop_binCRWS_muon2017_bin7
374 prop_binSR_electron2017_bin13_Triboson
375 prop_binCRTT_muon2018_bin7
376 prop_binCRWS_muon2016M_bin5_Fake
377 prop_binCRWS_electron2018_bin13
378 prop_binSR_electron2017_bin18_sm
379 prop_binSR_muon2016M_bin20_Triboson
380 prop_binSR_muon2016M_bin8_sm
381 prop_binSR_muon2017_bin10
382 prop_binSR_electron2018_bin18_ZZtoLep
383 prop_binSR_muon2016M_bin20_TVX
384 prop_binCRTT_electron2017_bin16
385 prop_binCRWS_muon2017_bin3
386 prop_binCRWS_electron2016M_bin6_Fake
387 prop_binCRWS_electron2017_bin14_Fake
388 prop_binCRTT_electron2016M_bin8
389 prop_binSR_muon2016M_bin17_sm
390 prop_binCRWS_electron2016M_bin19
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW



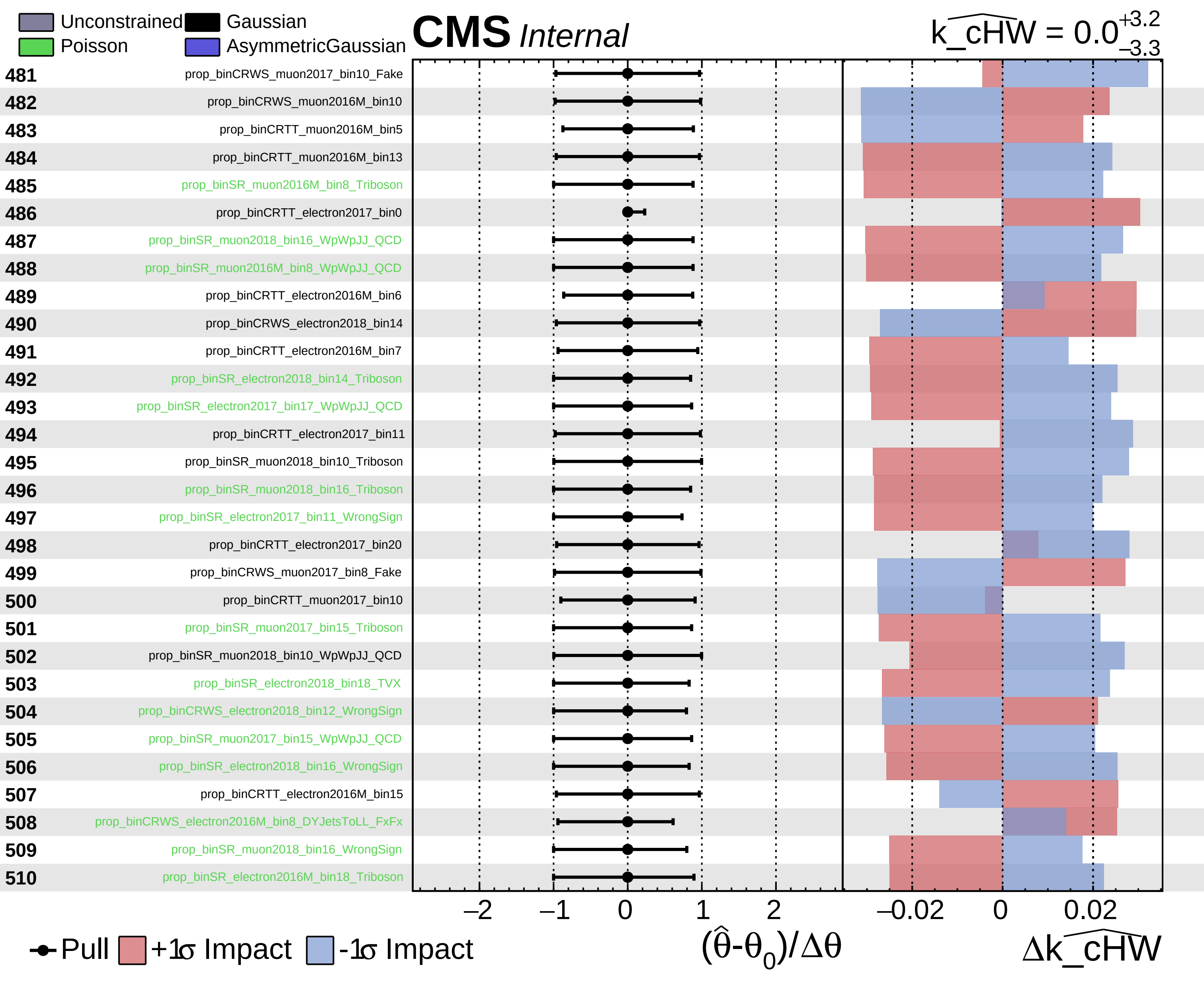
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+3.2}_{-3.3}$
391 prop_binSR_muon2016M_bin13_sm
392 prop_binCRTT_electron2018_bin15
393 prop_binCRWS_electron2017_bin14_TTTo2L2Nu
394 prop_binSR_electron2017_bin13_WpWpJJ_QCD
395 prop_binSR_muon2016M_bin13_TVX
396 prop_binSR_muon2016M_bin10_sm
397 prop_binSR_electron2016M_bin9_sm
398 prop_binSR_muon2016M_bin14_sm
399 prop_binSR_electron2017_bin9_WpWpJJ_QCD
400 prop_binCRTT_electron2016M_bin11
401 prop_binSR_muon2016M_bin8_TVX
402 prop_binSR_muon2017_bin12_Triboson
403 prop_binCRWS_electron2016M_bin12
404 prop_binSR_electron2016M_bin18_sm
405 prop_binSR_muon2016M_bin16_sm
406 prop_binCRTT_muon2016M_bin17
407 prop_binSR_muon2016M_bin15_sm
408 prop_binSR_electron2016M_bin16_sm
409 prop_binCRTT_electron2018_bin6
410 prop_binSR_electron2016M_bin20_Triboson
411 prop_binCRWS_electron2017_bin4
412 prop_binCRTT_electron2017_bin9
413 prop_binCRWS_electron2018_bin19_Fake
414 prop_binSR_muon2016M_bin14_ZZtoLep
415 prop_binCRTT_muon2016M_bin16
416 prop_binCRWS_muon2017_bin0
417 prop_binSR_electron2017_bin9_TVX
418 prop_binSR_muon2016M_bin12
419 prop_binCRTT_muon2017_bin8
420 prop_binSR_muon2017_bin11_WpWpJJ_QCD
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



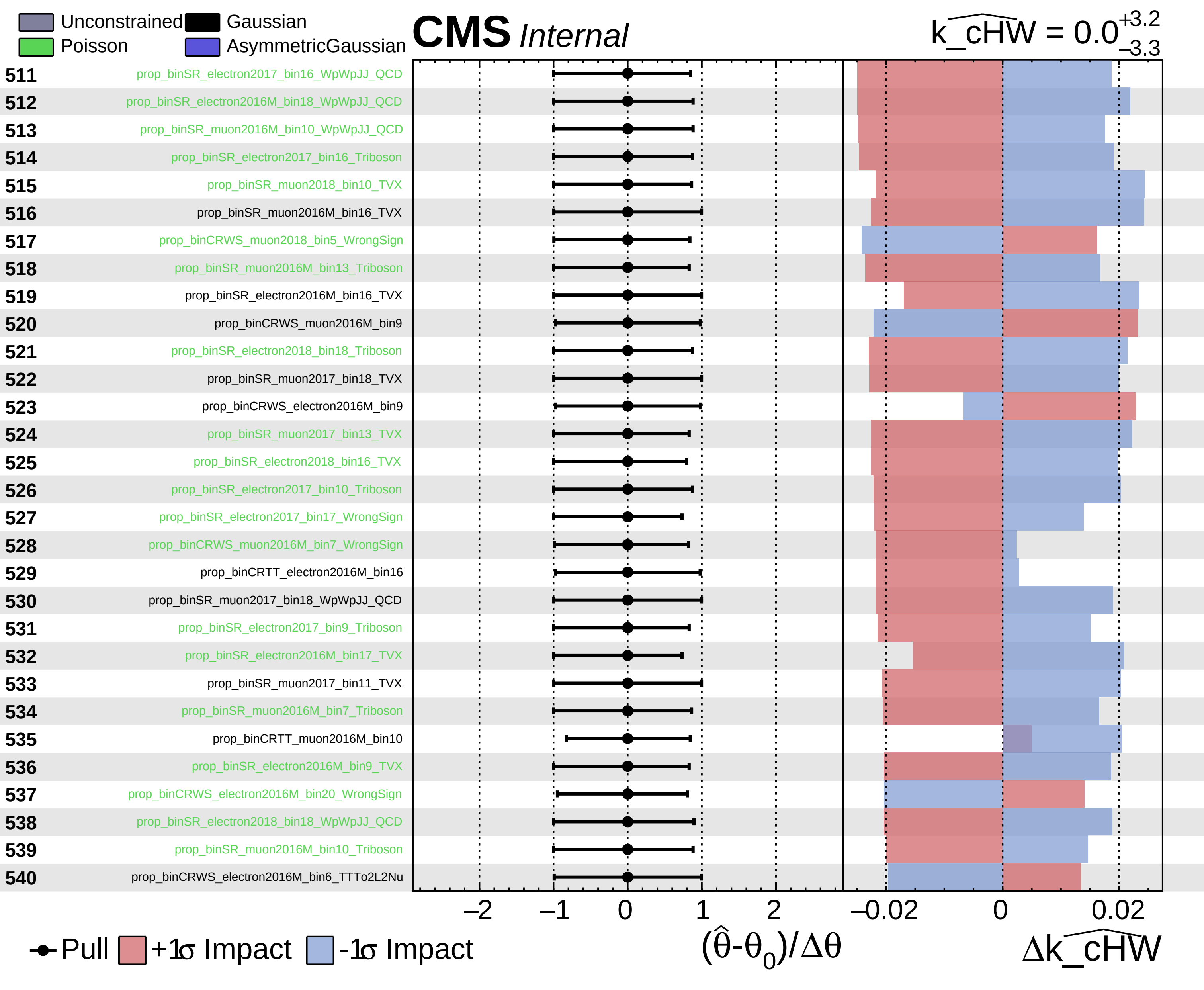
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+3.2}_{-3.3}$
421 prop_binSR_electron2016M_bin14
422 prop_binSR_electron2016M_bin10
423 prop_binCRWS_muon2016M_bin13
424 prop_binSR_electron2017_bin5_WpWpJJ_QCD
425 prop_binCRTT_electron2018_bin11
426 prop_binCRTT_electron2016M_bin5
427 prop_binCRWS_electron2017_bin5
428 prop_binCRWS_muon2016M_bin18
429 prop_binCRWS_muon2016M_bin6
430 prop_binSR_muon2017_bin11_WrongSign
431 prop_binSR_electron2017_bin16_WrongSign
432 prop_binCRWS_muon2018_bin6_DYJetsToLL_FxFx
433 prop_binCRTT_electron2018_bin8
434 prop_binCRWS_electron2018_bin12_TTTo2L2Nu
435 prop_binCRWS_electron2018_bin16
436 prop_binCRTT_electron2016M_bin12
437 prop_binCRTT_muon2017_bin12
438 prop_binCRWS_muon2018_bin13
439 prop_binCRTT_electron2017_bin15
440 prop_binCRTT_electron2016M_bin18
441 prop_binSR_electron2016M_bin12_sm
442 prop_binCRTT_electron2017_bin7
443 prop_binSR_muon2017_bin15_TVX
444 prop_binCRTT_electron2018_bin5
445 prop_binSR_muon2016M_bin11_sm
446 prop_binSR_muon2018_bin11_Triboson
447 prop_binCRWS_electron2016M_bin3
448 prop_binSR_electron2016M_bin18_TVX
449 prop_binCRTT_muon2016M_bin20
450 prop_binCRTT_electron2018_bin16
−2 −1 0 1 2
−0.05 0 0.05
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

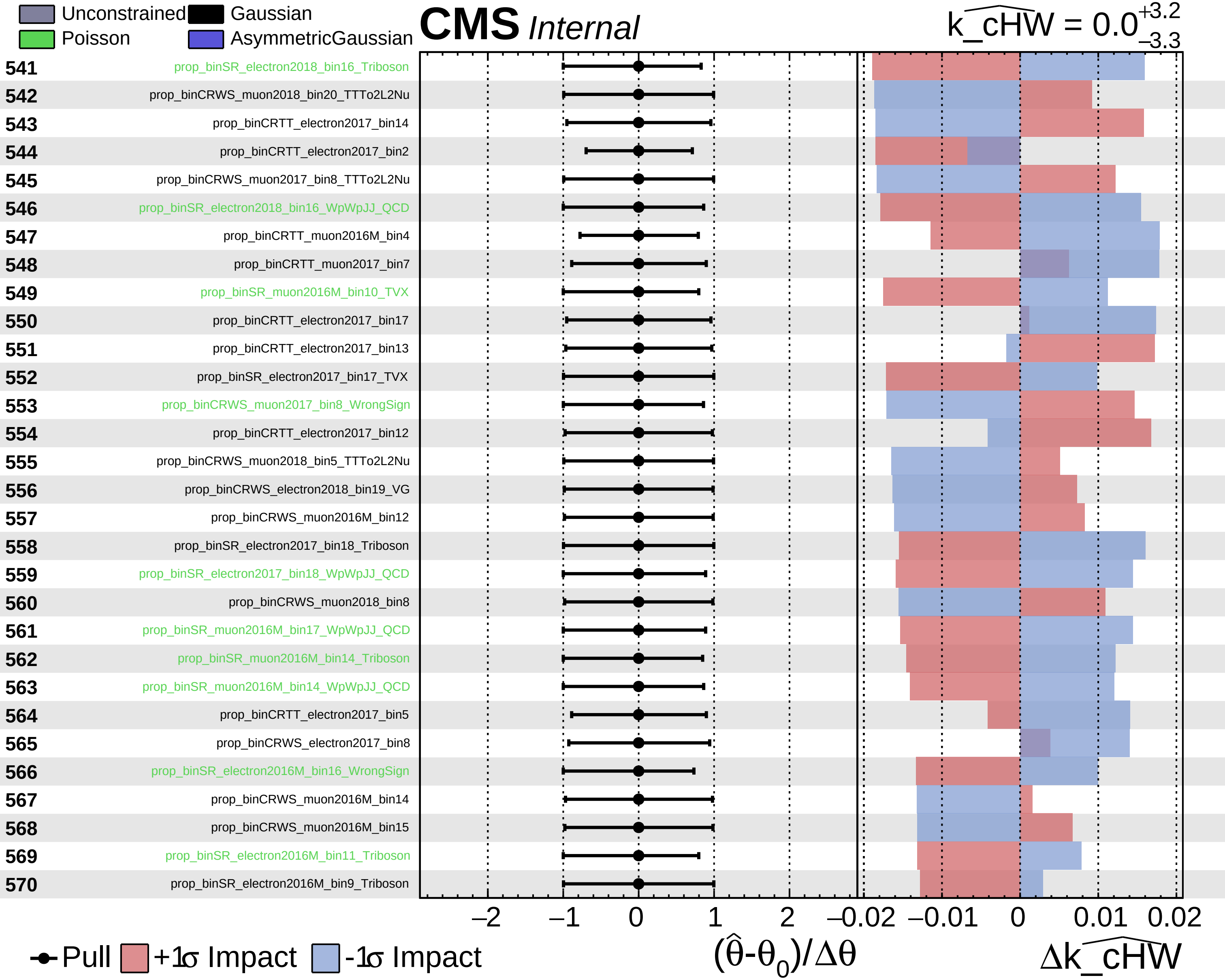


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
451 prop_binSR_electron2017_bin9_WrongSign
452 prop_binSR_electron2017_bin13_TVX
453 prop_binSR_muon2017_bin11_Triboson
454 prop_binSR_electron2017_bin11_WpWpJJ_QCD
455 prop_binSR_muon2016M_bin10_WrongSign
456 prop_binCRTT_muon2017_bin18
457 prop_binCRTT_electron2018_bin14
458 prop_binSR_muon2016M_bin7_WrongSign
459 prop_binCRTT_muon2016M_bin1
460 prop_binSR_muon2017_bin15_WrongSign
461 prop_binSR_muon2017_bin9_sm
462 prop_binCRTT_muon2016M_bin14
463 prop_binSR_muon2016M_bin7_WpWpJJ_QCD
464 prop_binSR_electron2016M_bin11_TVX
465 prop_binSR_electron2016M_bin11_WpWpJJ_QCD
466 prop_binCRTT_muon2018_bin10
467 prop_binSR_electron2017_bin5_Triboson
468 prop_binSR_electron2017_bin10_WpWpJJ_QCD
469 prop_binCRWS_electron2016M_bin6_WrongSign
470 prop_binSR_electron2017_bin11_TVX
471 prop_binSR_muon2016M_bin13_WpWpJJ_QCD
472 lep_2016M
473 prop_binCRTT_muon2018_bin18
474 prop_binSR_muon2018_bin16_TVX
475 prop_binSR_electron2018_bin14_WpWpJJ_QCD
476 prop_binCRWS_electron2017_bin18
477 prop_binCRTT_muon2018_bin17
478 prop_binSR_electron2017_bin5_WrongSign
479 prop_binCRWS_electron2017_bin14_WrongSign
480 prop_binSR_electron2018_bin14_TVX
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

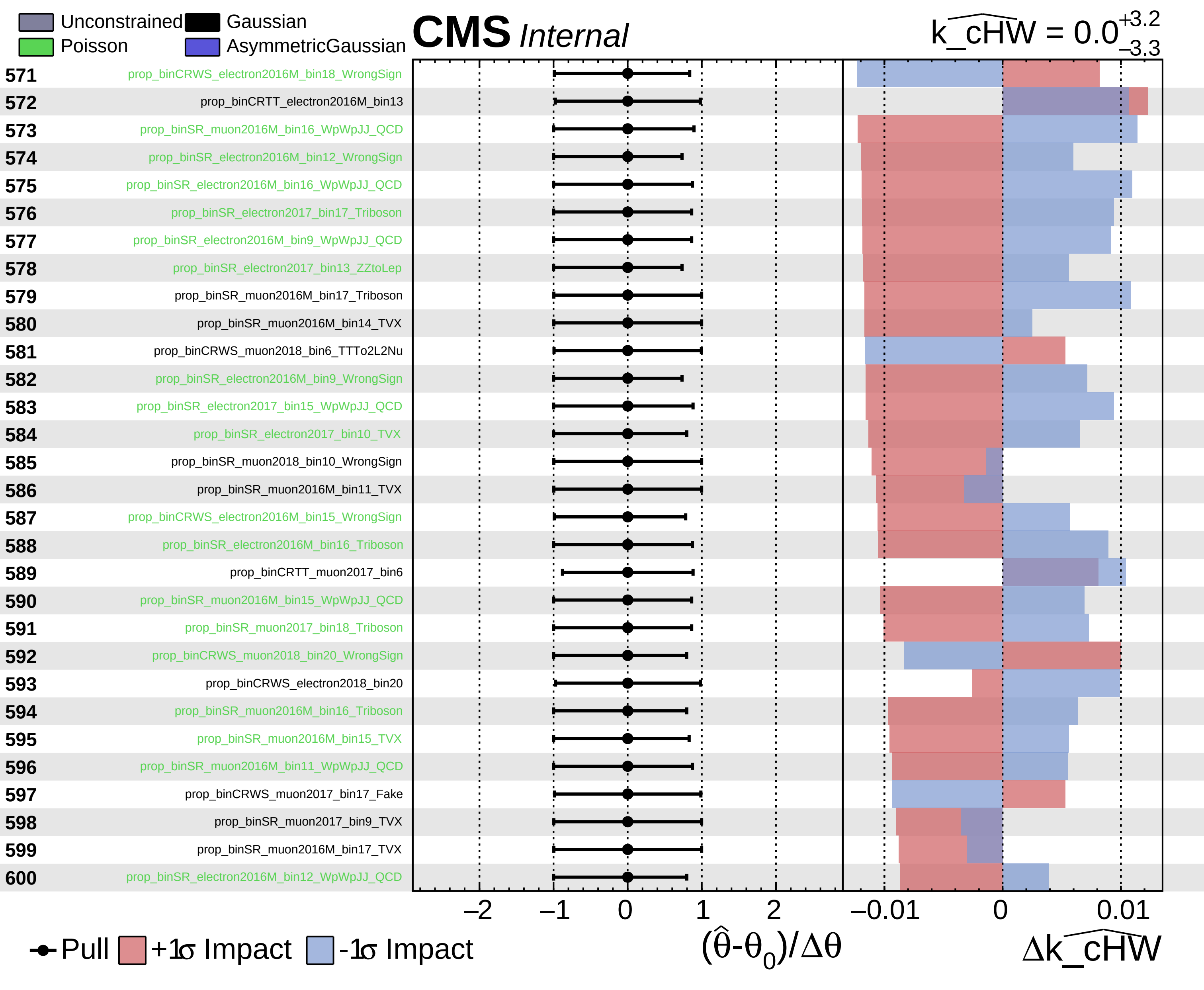


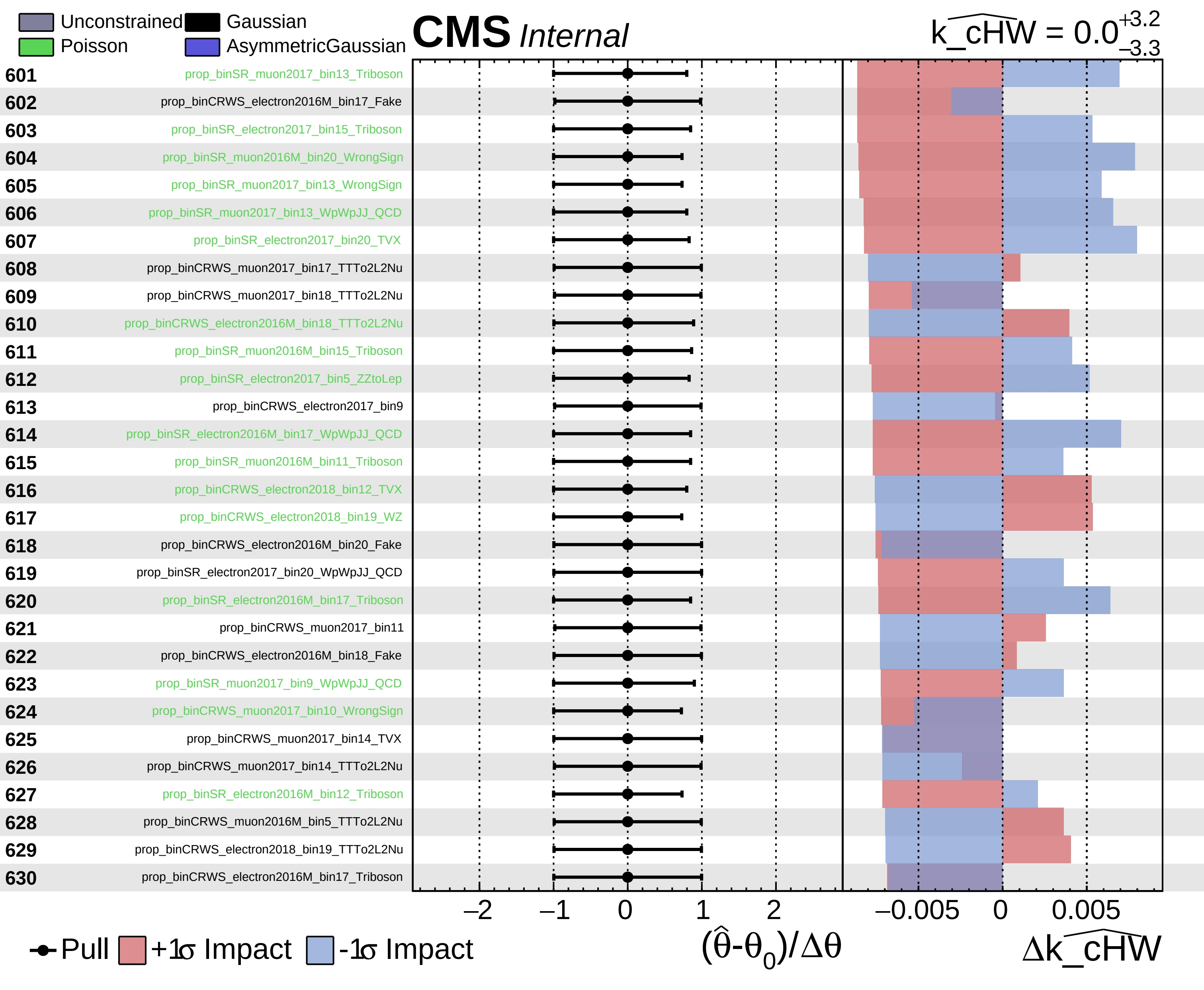
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
481 prop_binCRWS_muon2017_bin10_Fake
482 prop_binCRWS_muon2016M_bin10
483 prop_binCRTT_muon2016M_bin5
484 prop_binCRTT_muon2016M_bin13
485 prop_binSR_muon2016M_bin8_Triboson
486 prop_binCRTT_electron2017_bin0
487 prop_binSR_muon2018_bin16_WpWpJJ_QCD
488 prop_binSR_muon2016M_bin8_WpWpJJ_QCD
489 prop_binCRTT_electron2016M_bin6
490 prop_binCRWS_electron2018_bin14
491 prop_binCRTT_electron2016M_bin7
492 prop_binSR_electron2018_bin14_Triboson
493 prop_binSR_electron2017_bin17_WpWpJJ_QCD
494 prop_binCRTT_electron2017_bin11
495 prop_binSR_muon2018_bin10_Triboson
496 prop_binSR_muon2018_bin16_Triboson
497 prop_binSR_electron2017_bin11_WrongSign
498 prop_binCRTT_electron2017_bin20
499 prop_binCRWS_muon2017_bin8_Fake
500 prop_binCRTT_muon2017_bin10
501 prop_binSR_muon2017_bin15_Triboson
502 prop_binSR_muon2018_bin10_WpWpJJ_QCD
503 prop_binSR_electron2018_bin18_TVX
504 prop_binCRWS_electron2018_bin12_WrongSign
505 prop_binSR_muon2017_bin15_WpWpJJ_QCD
506 prop_binSR_electron2018_bin16_WrongSign
507 prop_binCRTT_electron2016M_bin15
508 prop_binCRWS_electron2016M_bin8_DYJetsToLL_FxFx
509 prop_binSR_muon2018_bin16_WrongSign
510 prop_binSR_electron2016M_bin18_Triboson
−2 −1 0 1 2
−0.02 0 0.02
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
511 prop_binSR_electron2017_bin16_WpWpJJ_QCD
512 prop_binSR_electron2016M_bin18_WpWpJJ_QCD
513 prop_binSR_muon2016M_bin10_WpWpJJ_QCD
514 prop_binSR_electron2017_bin16_Triboson
515 prop_binSR_muon2018_bin10_TVX
516 prop_binSR_muon2016M_bin16_TVX
517 prop_binCRWS_muon2018_bin5_WrongSign
518 prop_binSR_muon2016M_bin13_Triboson
519 prop_binSR_electron2016M_bin16_TVX
520 prop_binCRWS_muon2016M_bin9
521 prop_binSR_electron2018_bin18_Triboson
522 prop_binSR_muon2017_bin18_TVX
523 prop_binCRWS_electron2016M_bin9
524 prop_binSR_muon2017_bin13_TVX
525 prop_binSR_electron2018_bin16_TVX
526 prop_binSR_electron2017_bin10_Triboson
527 prop_binSR_electron2017_bin17_WrongSign
528 prop_binCRWS_muon2016M_bin7_WrongSign
529 prop_binCRTT_electron2016M_bin16
530 prop_binSR_muon2017_bin18_WpWpJJ_QCD
531 prop_binSR_electron2017_bin9_Triboson
532 prop_binSR_electron2016M_bin17_TVX
533 prop_binSR_muon2017_bin11_TVX
534 prop_binSR_muon2016M_bin7_Triboson
535 prop_binCRTT_muon2016M_bin10
536 prop_binSR_electron2016M_bin9_TVX
537 prop_binCRWS_electron2016M_bin20_WrongSign
538 prop_binSR_electron2018_bin18_WpWpJJ_QCD
539 prop_binSR_muon2016M_bin10_Triboson
540 prop_binCRWS_electron2016M_bin6_TTTo2L2Nu
−2 −1 0 1 2
−0.02 0 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

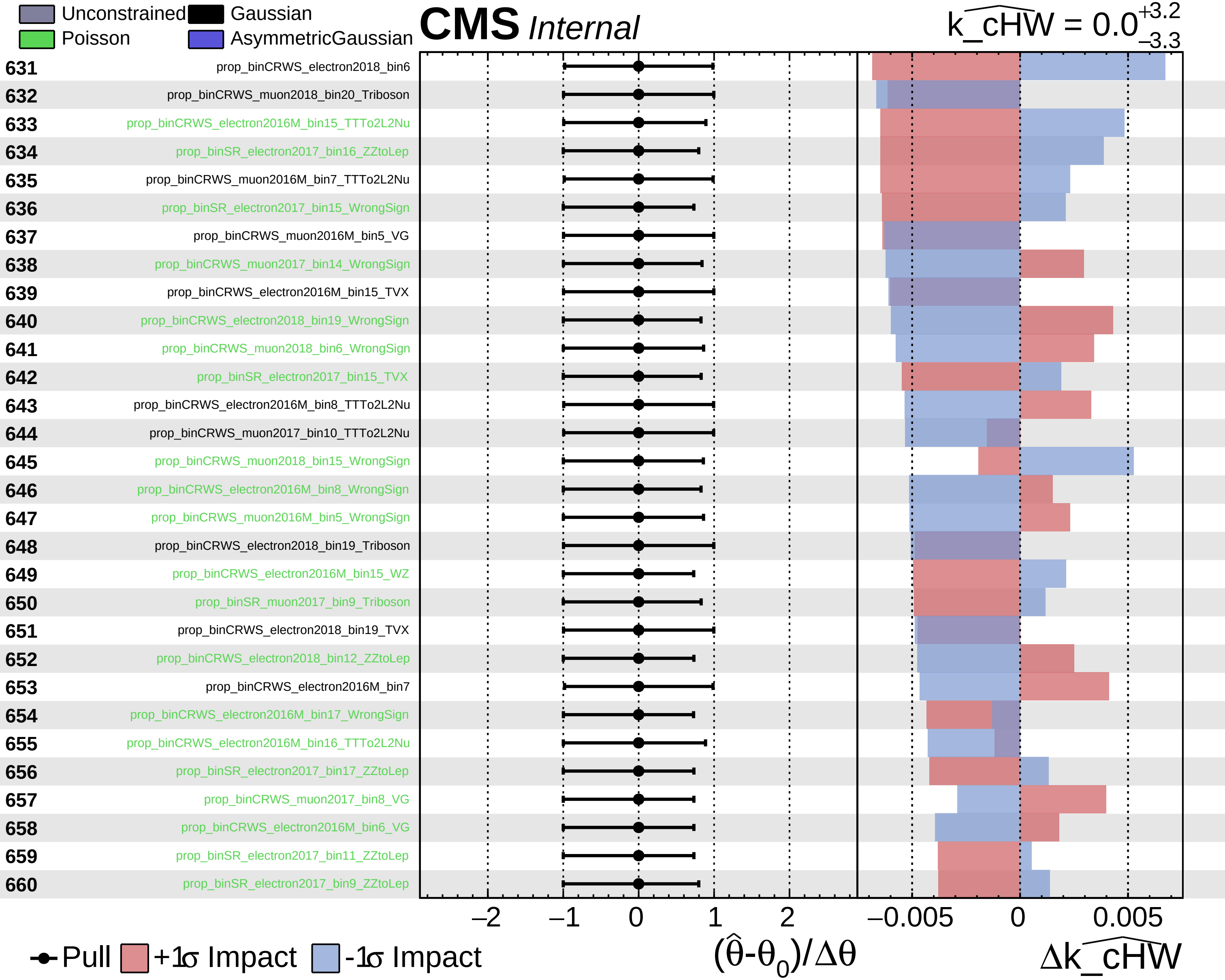


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
541 prop_binSR_electron2018_bin16_Triboson
542 prop_binCRWS_muon2018_bin20_TTTo2L2Nu
543 prop_binCRTT_electron2017_bin14
544 prop_binCRTT_electron2017_bin2
545 prop_binCRWS_muon2017_bin8_TTTo2L2Nu
546 prop_binSR_electron2018_bin16_WpWpJJ_QCD
547 prop_binCRTT_muon2016M_bin4
548 prop_binCRTT_muon2017_bin7
549 prop_binSR_muon2016M_bin10_TVX
550 prop_binCRTT_electron2017_bin17
551 prop_binCRTT_electron2017_bin13
552 prop_binSR_electron2017_bin17_TVX
553 prop_binCRWS_muon2017_bin8_WrongSign
554 prop_binCRTT_electron2017_bin12
555 prop_binCRWS_muon2018_bin5_TTTo2L2Nu
556 prop_binCRWS_electron2018_bin19_VG
557 prop_binCRWS_muon2016M_bin12
558 prop_binSR_electron2017_bin18_Triboson
559 prop_binSR_electron2017_bin18_WpWpJJ_QCD
560 prop_binCRWS_muon2018_bin8
561 prop_binSR_muon2016M_bin17_WpWpJJ_QCD
562 prop_binSR_muon2016M_bin14_Triboson
563 prop_binSR_muon2016M_bin14_WpWpJJ_QCD
564 prop_binCRTT_electron2017_bin5
565 prop_binCRWS_electron2017_bin8
566 prop_binSR_electron2016M_bin16_WrongSign
567 prop_binCRWS_muon2016M_bin14
568 prop_binCRWS_muon2016M_bin15
569 prop_binSR_electron2016M_bin11_Triboson
570 prop_binSR_electron2016M_bin9_Triboson
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

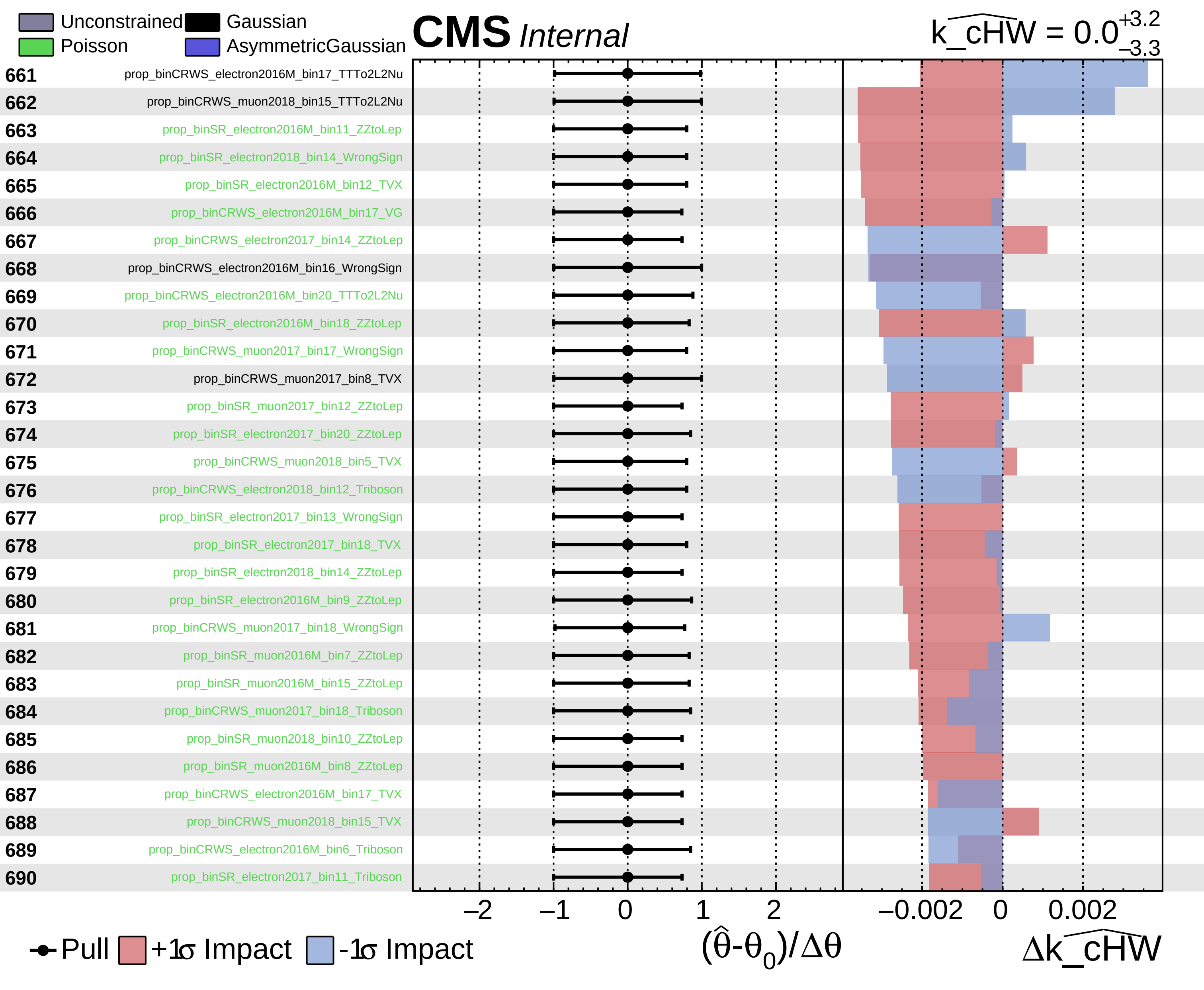

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
571 prop_binCRWS_electron2016M_bin18_WrongSign
572 prop_binCRTT_electron2016M_bin13
573 prop_binSR_muon2016M_bin16_WpWpJJ_QCD
574 prop_binSR_electron2016M_bin12_WrongSign
575 prop_binSR_electron2016M_bin16_WpWpJJ_QCD
576 prop_binSR_electron2017_bin17_Triboson
577 prop_binSR_electron2016M_bin9_WpWpJJ_QCD
578 prop_binSR_electron2017_bin13_ZZtoLep
579 prop_binSR_muon2016M_bin17_Triboson
580 prop_binSR_muon2016M_bin14_TVX
581 prop_binCRWS_muon2018_bin6_TTTo2L2Nu
582 prop_binSR_electron2016M_bin9_WrongSign
583 prop_binSR_electron2017_bin15_WpWpJJ_QCD
584 prop_binSR_electron2017_bin10_TVX
585 prop_binSR_muon2018_bin10_WrongSign
586 prop_binSR_muon2016M_bin11_TVX
587 prop_binCRWS_electron2016M_bin15_WrongSign
588 prop_binSR_electron2016M_bin16_Triboson
589 prop_binCRTT_muon2017_bin6
590 prop_binSR_muon2016M_bin15_WpWpJJ_QCD
591 prop_binSR_muon2017_bin18_Triboson
592 prop_binCRWS_muon2018_bin20_WrongSign
593 prop_binCRWS_electron2018_bin20
594 prop_binSR_muon2016M_bin16_Triboson
595 prop_binSR_muon2016M_bin15_TVX
596 prop_binSR_muon2016M_bin11_WpWpJJ_QCD
597 prop_binCRWS_muon2017_bin17_Fake
598 prop_binSR_muon2017_bin9_TVX
599 prop_binSR_muon2016M_bin17_TVX
600 prop_binSR_electron2016M_bin12_WpWpJJ_QCD
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

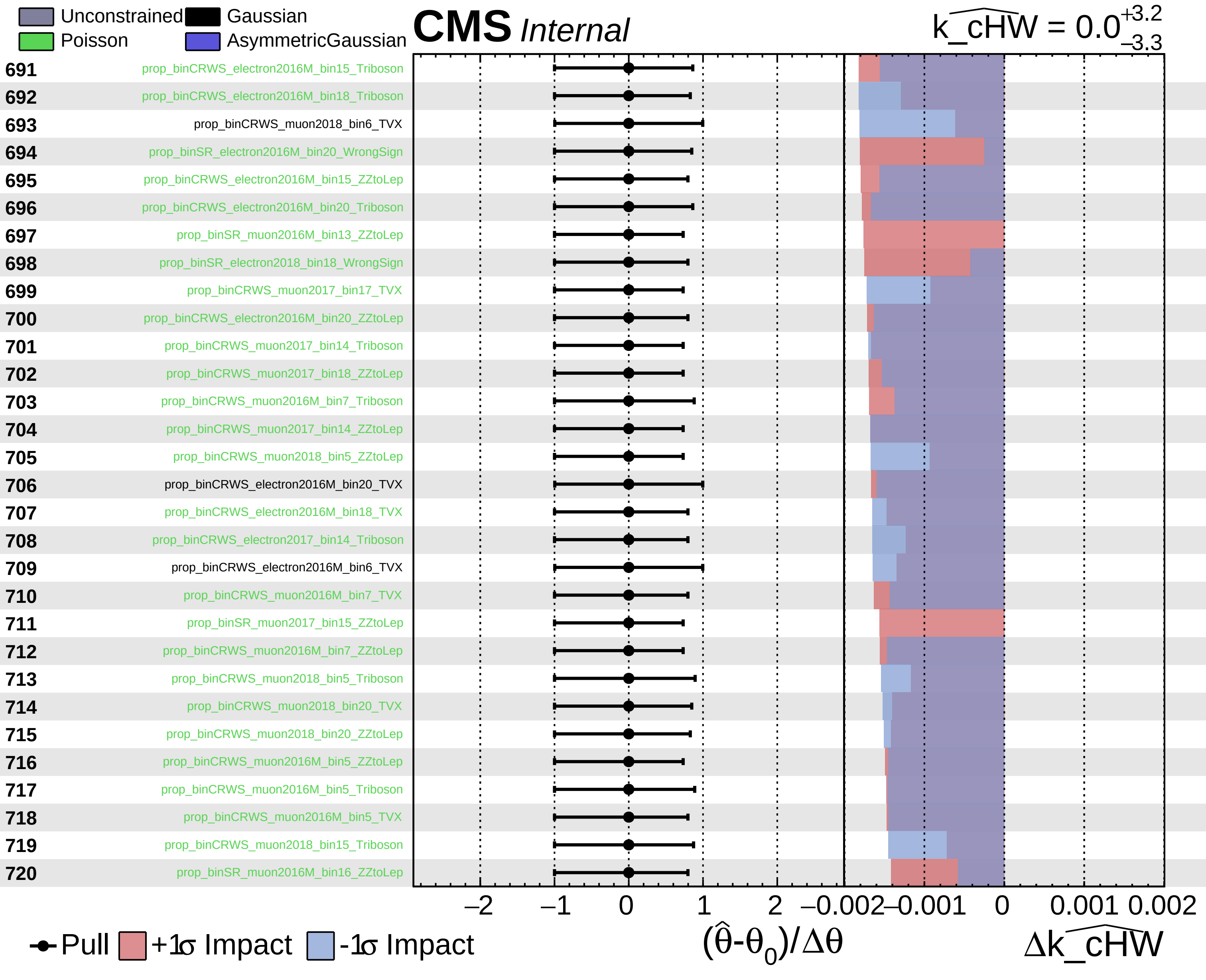


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
601 prop_binSR_muon2017_bin13_Triboson
602 prop_binCRWS_electron2016M_bin17_Fake
603 prop_binSR_electron2017_bin15_Triboson
604 prop_binSR_muon2016M_bin20_WrongSign
605 prop_binSR_muon2017_bin13_WrongSign
606 prop_binSR_muon2017_bin13_WpWpJJ_QCD
607 prop_binSR_electron2017_bin20_TVX
608 prop_binCRWS_muon2017_bin17_TTTo2L2Nu
609 prop_binCRWS_muon2017_bin18_TTTo2L2Nu
610 prop_binCRWS_electron2016M_bin18_TTTo2L2Nu
611 prop_binSR_muon2016M_bin15_Triboson
612 prop_binSR_electron2017_bin5_ZZtoLep
613 prop_binCRWS_electron2017_bin9
614 prop_binSR_electron2016M_bin17_WpWpJJ_QCD
615 prop_binSR_muon2016M_bin11_Triboson
616 prop_binCRWS_electron2018_bin12_TVX
617 prop_binCRWS_electron2018_bin19_WZ
618 prop_binCRWS_electron2016M_bin20_Fake
619 prop_binSR_electron2017_bin20_WpWpJJ_QCD
620 prop_binSR_electron2016M_bin17_Triboson
621 prop_binCRWS_muon2017_bin11
622 prop_binCRWS_electron2016M_bin18_Fake
623 prop_binSR_muon2017_bin9_WpWpJJ_QCD
624 prop_binCRWS_muon2017_bin10_WrongSign
625 prop_binCRWS_muon2017_bin14_TVX
626 prop_binCRWS_muon2017_bin14_TTTo2L2Nu
627 prop_binSR_electron2016M_bin12_Triboson
628 prop_binCRWS_muon2016M_bin5_TTTo2L2Nu
629 prop_binCRWS_electron2018_bin19_TTTo2L2Nu
630 prop_binCRWS_electron2016M_bin17_Triboson
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

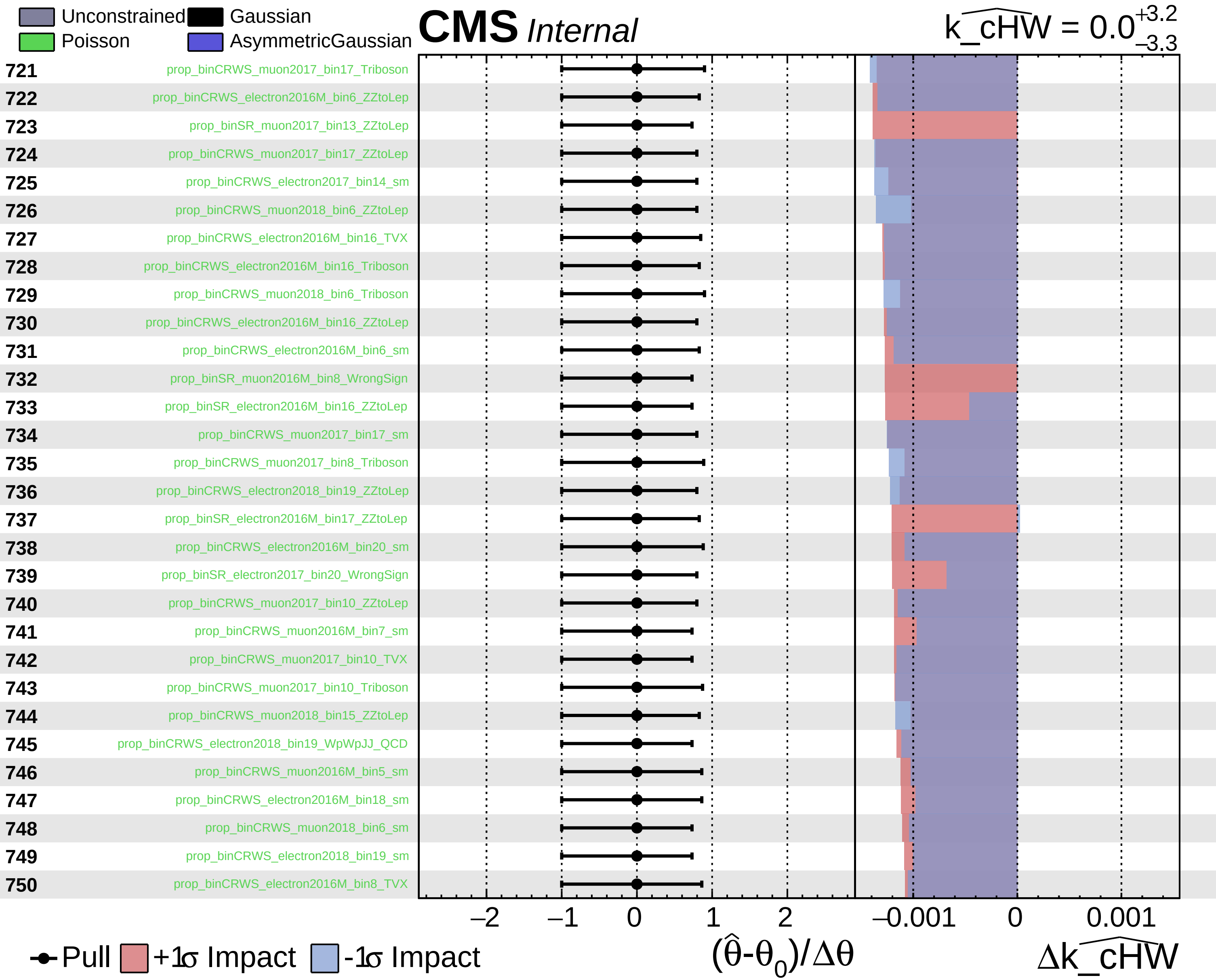


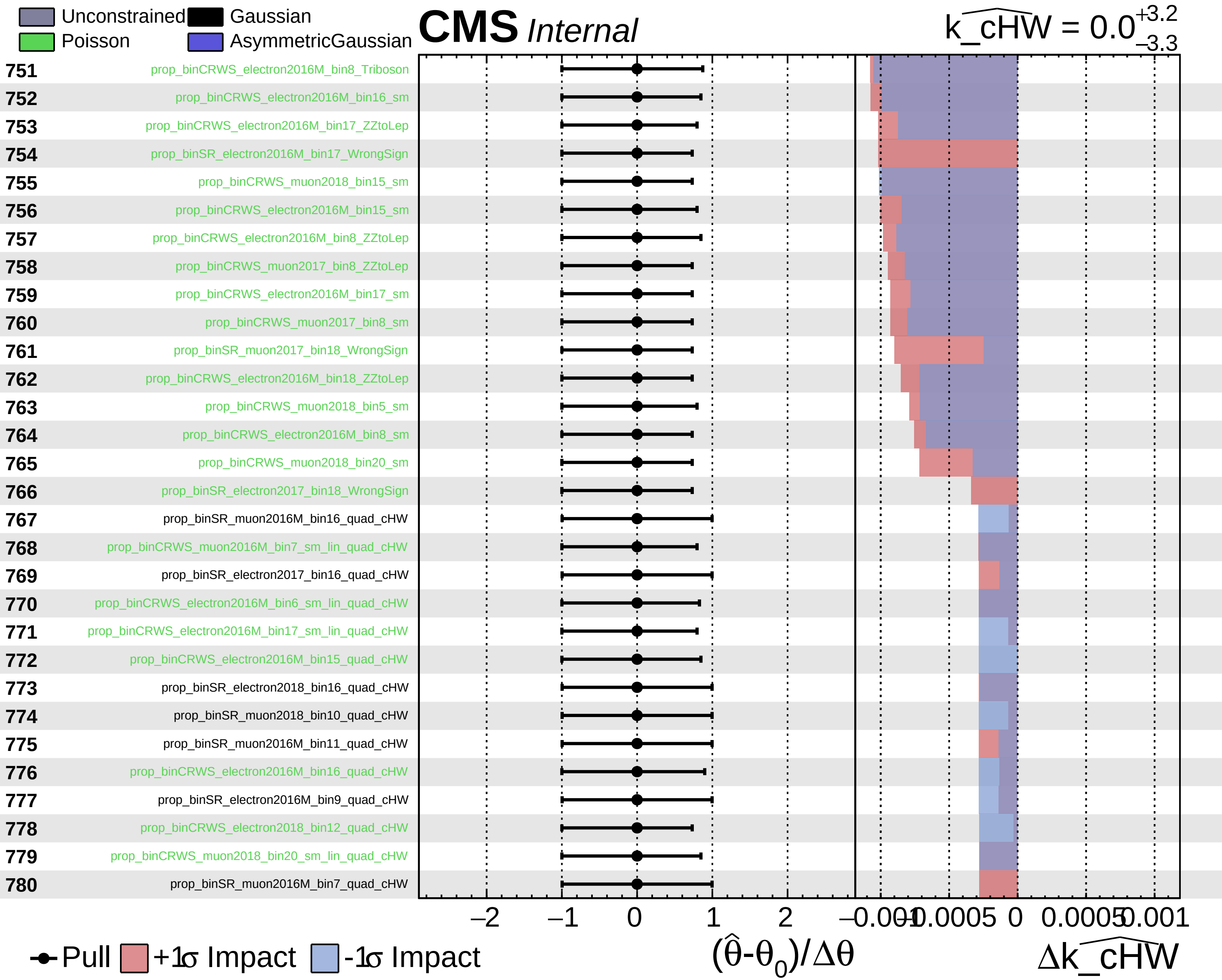
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
631 prop_binCRWS_electron2018_bin6
632 prop_binCRWS_muon2018_bin20_Triboson
633 prop_binCRWS_electron2016M_bin15_TTTo2L2Nu
634 prop_binSR_electron2017_bin16_ZZtoLep
635 prop_binCRWS_muon2016M_bin7_TTTo2L2Nu
636 prop_binSR_electron2017_bin15_WrongSign
637 prop_binCRWS_muon2016M_bin5_VG
638 prop_binCRWS_muon2017_bin14_WrongSign
639 prop_binCRWS_electron2016M_bin15_TVX
640 prop_binCRWS_electron2018_bin19_WrongSign
641 prop_binCRWS_muon2018_bin6_WrongSign
642 prop_binSR_electron2017_bin15_TVX
643 prop_binCRWS_electron2016M_bin8_TTTo2L2Nu
644 prop_binCRWS_muon2017_bin10_TTTo2L2Nu
645 prop_binCRWS_muon2018_bin15_WrongSign
646 prop_binCRWS_electron2016M_bin8_WrongSign
647 prop_binCRWS_muon2016M_bin5_WrongSign
648 prop_binCRWS_electron2018_bin19_Triboson
649 prop_binCRWS_electron2016M_bin15_WZ
650 prop_binSR_muon2017_bin9_Triboson
651 prop_binCRWS_electron2018_bin19_TVX
652 prop_binCRWS_electron2018_bin12_ZZtoLep
653 prop_binCRWS_electron2016M_bin7
654 prop_binCRWS_electron2016M_bin17_WrongSign
655 prop_binCRWS_electron2016M_bin16_TTTo2L2Nu
656 prop_binSR_electron2017_bin17_ZZtoLep
657 prop_binCRWS_muon2017_bin8_VG
658 prop_binCRWS_electron2016M_bin6_VG
659 prop_binSR_electron2017_bin11_ZZtoLep
660 prop_binSR_electron2017_bin9_ZZtoLep
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

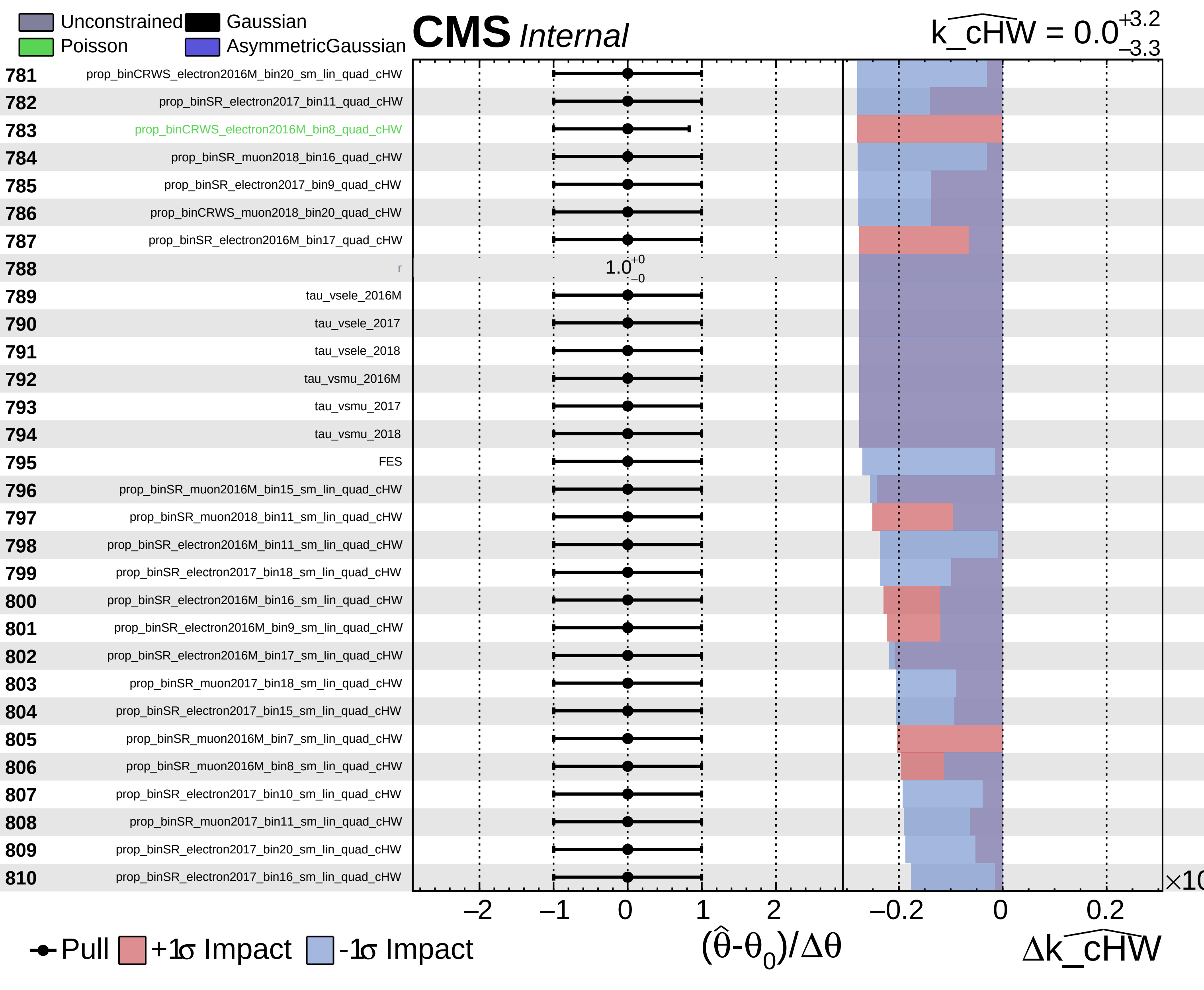

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+3.2}_{-3.3}$
661 prop_binCRWS_electron2016M_bin17_TTTo2L2Nu
662 prop_binCRWS_muon2018_bin15_TTTo2L2Nu
663 prop_binSR_electron2016M_bin11_ZZtoLep
664 prop_binSR_electron2018_bin14_WrongSign
665 prop_binSR_electron2016M_bin12_TVX
666 prop_binCRWS_electron2016M_bin17_VG
667 prop_binCRWS_electron2017_bin14_ZZtoLep
668 prop_binCRWS_electron2016M_bin16_WrongSign
669 prop_binCRWS_electron2016M_bin20_TTTo2L2Nu
670 prop_binSR_electron2016M_bin18_ZZtoLep
671 prop_binCRWS_muon2017_bin17_WrongSign
672 prop_binCRWS_muon2017_bin8_TVX
673 prop_binSR_muon2017_bin12_ZZtoLep
674 prop_binSR_electron2017_bin20_ZZtoLep
675 prop_binCRWS_muon2018_bin5_TVX
676 prop_binCRWS_electron2018_bin12_Triboson
677 prop_binSR_electron2017_bin13_WrongSign
678 prop_binSR_electron2017_bin18_TVX
679 prop_binSR_electron2018_bin14_ZZtoLep
680 prop_binSR_electron2016M_bin9_ZZtoLep
681 prop_binCRWS_muon2017_bin18_WrongSign
682 prop_binSR_muon2016M_bin7_ZZtoLep
683 prop_binSR_muon2016M_bin15_ZZtoLep
684 prop_binCRWS_muon2017_bin18_Triboson
685 prop_binSR_muon2018_bin10_ZZtoLep
686 prop_binSR_muon2016M_bin8_ZZtoLep
687 prop_binCRWS_electron2016M_bin17_TVX
688 prop_binCRWS_muon2018_bin15_TVX
689 prop_binCRWS_electron2016M_bin6_Triboson
690 prop_binSR_electron2017_bin11_Triboson
−2
−1
0
1
2
−0.002
0
0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$



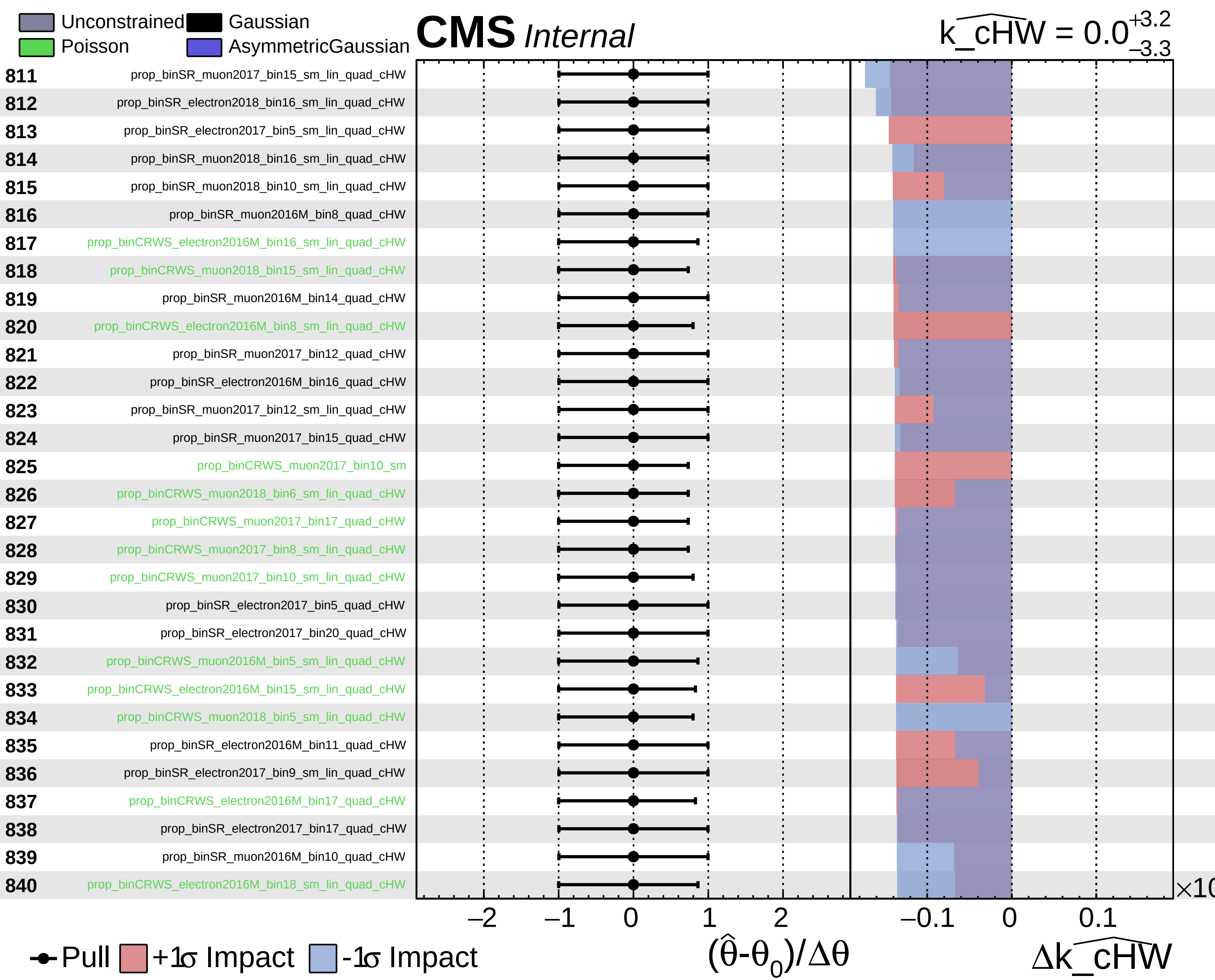
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = $0.0^{+3.2}_{-3.3}$
691 prop_binCRWS_electron2016M_bin15_Triboson
692 prop_binCRWS_electron2016M_bin18_Triboson
693 prop_binCRWS_muon2018_bin6_TVX
694 prop_binSR_electron2016M_bin20_WrongSign
695 prop_binCRWS_electron2016M_bin15_ZZtoLep
696 prop_binCRWS_electron2016M_bin20_Triboson
697 prop_binSR_muon2016M_bin13_ZZtoLep
698 prop_binSR_electron2018_bin18_WrongSign
699 prop_binCRWS_muon2017_bin17_TVX
700 prop_binCRWS_electron2016M_bin20_ZZtoLep
701 prop_binCRWS_muon2017_bin14_Triboson
702 prop_binCRWS_muon2017_bin18_ZZtoLep
703 prop_binCRWS_muon2016M_bin7_Triboson
704 prop_binCRWS_muon2017_bin14_ZZtoLep
705 prop_binCRWS_muon2018_bin5_ZZtoLep
706 prop_binCRWS_electron2016M_bin20_TVX
707 prop_binCRWS_electron2016M_bin18_TVX
708 prop_binCRWS_electron2017_bin14_Triboson
709 prop_binCRWS_electron2016M_bin6_TVX
710 prop_binCRWS_muon2016M_bin7_TVX
711 prop_binSR_muon2017_bin15_ZZtoLep
712 prop_binCRWS_muon2016M_bin7_ZZtoLep
713 prop_binCRWS_muon2018_bin5_Triboson
714 prop_binCRWS_muon2018_bin20_TVX
715 prop_binCRWS_muon2018_bin20_ZZtoLep
716 prop_binCRWS_muon2016M_bin5_ZZtoLep
717 prop_binCRWS_muon2016M_bin5_Triboson
718 prop_binCRWS_muon2016M_bin5_TVX
719 prop_binCRWS_muon2018_bin15_Triboson
720 prop_binSR_muon2016M_bin16_ZZtoLep
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
Δk_cHW


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+3.2}_{-3.3}$
721 prop_binCRWS_muon2017_bin17_Triboson
722 prop_binCRWS_electron2016M_bin6_ZZtoLep
723 prop_binSR_muon2017_bin13_ZZtoLep
724 prop_binCRWS_muon2017_bin17_ZZtoLep
725 prop_binCRWS_electron2017_bin14_sm
726 prop_binCRWS_muon2018_bin6_ZZtoLep
727 prop_binCRWS_electron2016M_bin16_TVX
728 prop_binCRWS_electron2016M_bin16_Triboson
729 prop_binCRWS_muon2018_bin6_Triboson
730 prop_binCRWS_electron2016M_bin16_ZZtoLep
731 prop_binCRWS_electron2016M_bin6_sm
732 prop_binSR_muon2016M_bin8_WrongSign
733 prop_binSR_electron2016M_bin16_ZZtoLep
734 prop_binCRWS_muon2017_bin17_sm
735 prop_binCRWS_muon2017_bin8_Triboson
736 prop_binCRWS_electron2018_bin19_ZZtoLep
737 prop_binSR_electron2016M_bin17_ZZtoLep
738 prop_binCRWS_electron2016M_bin20_sm
739 prop_binSR_electron2017_bin20_WrongSign
740 prop_binCRWS_muon2017_bin10_ZZtoLep
741 prop_binCRWS_muon2016M_bin7_sm
742 prop_binCRWS_muon2017_bin10_TVX
743 prop_binCRWS_muon2017_bin10_Triboson
744 prop_binCRWS_muon2018_bin15_ZZtoLep
745 prop_binCRWS_electron2018_bin19_WpWpJJ_QCD
746 prop_binCRWS_muon2016M_bin5_sm
747 prop_binCRWS_electron2016M_bin18_sm
748 prop_binCRWS_muon2018_bin6_sm
749 prop_binCRWS_electron2018_bin19_sm
750 prop_binCRWS_electron2016M_bin8_TVX
−2 −1 0 1 2
−0.001 0 0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$

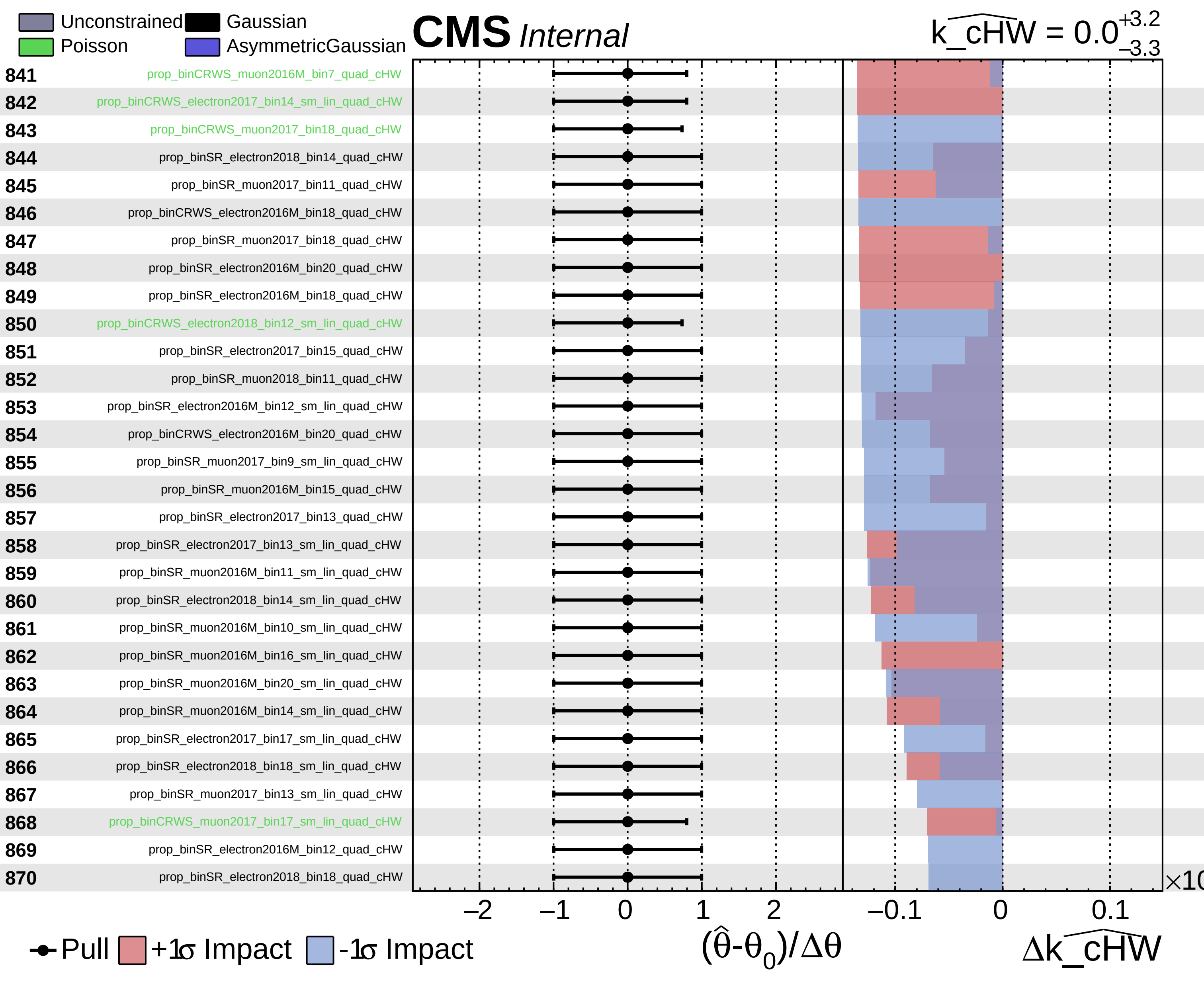

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
751 prop_binCRWS_electron2016M_bin8_Triboson
752 prop_binCRWS_electron2016M_bin16_sm
753 prop_binCRWS_electron2016M_bin17_ZZtoLep
754 prop_binSR_electron2016M_bin17_WrongSign
755 prop_binCRWS_muon2018_bin15_sm
756 prop_binCRWS_electron2016M_bin15_sm
757 prop_binCRWS_electron2016M_bin8_ZZtoLep
758 prop_binCRWS_muon2017_bin8_ZZtoLep
759 prop_binCRWS_electron2016M_bin17_sm
760 prop_binCRWS_muon2017_bin8_sm
761 prop_binSR_muon2017_bin18_WrongSign
762 prop_binCRWS_electron2016M_bin18_ZZtoLep
763 prop_binCRWS_muon2018_bin5_sm
764 prop_binCRWS_electron2016M_bin8_sm
765 prop_binCRWS_muon2018_bin20_sm
766 prop_binSR_electron2017_bin18_WrongSign
767 prop_binSR_muon2016M_bin16_quad_cHW
768 prop_binCRWS_muon2016M_bin7_sm_lin_quad_cHW
769 prop_binSR_electron2017_bin16_quad_cHW
770 prop_binCRWS_electron2016M_bin6_sm_lin_quad_cHW
771 prop_binCRWS_electron2016M_bin17_sm_lin_quad_cHW
772 prop_binCRWS_electron2016M_bin15_quad_cHW
773 prop_binSR_electron2018_bin16_quad_cHW
774 prop_binSR_muon2018_bin10_quad_cHW
775 prop_binSR_muon2016M_bin11_quad_cHW
776 prop_binCRWS_electron2016M_bin16_quad_cHW
777 prop_binSR_electron2016M_bin9_quad_cHW
778 prop_binCRWS_electron2018_bin12_quad_cHW
779 prop_binCRWS_muon2018_bin20_sm_lin_quad_cHW
780 prop_binSR_muon2016M_bin7_quad_cHW
−2
−1
0
1
2
−0.001
−0.0005
0
0.0005
0.001
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW

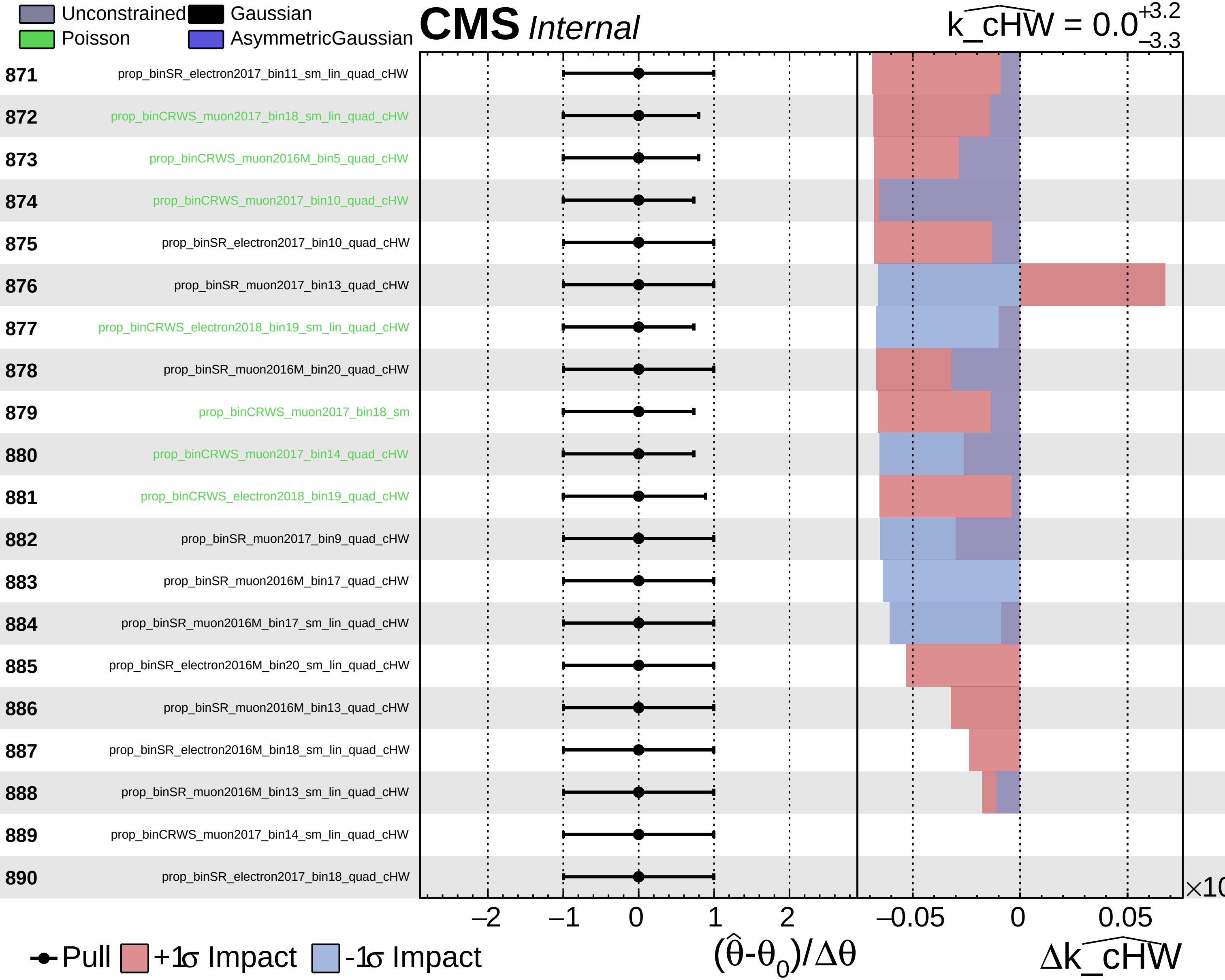


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
781 prop_binCRWS_electron2016M_bin20_sm_lin_quad_cHW
782 prop_binSR_electron2017_bin11_quad_cHW
783 prop_binCRWS_electron2016M_bin8_quad_cHW
784 prop_binSR_muon2018_bin16_quad_cHW
785 prop_binSR_electron2017_bin9_quad_cHW
786 prop_binCRWS_muon2018_bin20_quad_cHW
787 prop_binSR_electron2016M_bin17_quad_cHW
788 r
1.0 +0 −0
789 tau_vsele_2016M
790 tau_vsele_2017
791 tau_vsele_2018
792 tau_vsmu_2016M
793 tau_vsmu_2017
794 tau_vsmu_2018
795 FES
796 prop_binSR_muon2016M_bin15_sm_lin_quad_cHW
797 prop_binSR_muon2018_bin11_sm_lin_quad_cHW
798 prop_binSR_electron2016M_bin11_sm_lin_quad_cHW
799 prop_binSR_electron2017_bin18_sm_lin_quad_cHW
800 prop_binSR_electron2016M_bin16_sm_lin_quad_cHW
801 prop_binSR_electron2016M_bin9_sm_lin_quad_cHW
802 prop_binSR_electron2016M_bin17_sm_lin_quad_cHW
803 prop_binSR_muon2017_bin18_sm_lin_quad_cHW
804 prop_binSR_electron2017_bin15_sm_lin_quad_cHW
805 prop_binSR_muon2016M_bin7_sm_lin_quad_cHW
806 prop_binSR_muon2016M_bin8_sm_lin_quad_cHW
807 prop_binSR_electron2017_bin10_sm_lin_quad_cHW
808 prop_binSR_muon2017_bin11_sm_lin_quad_cHW
809 prop_binSR_electron2017_bin20_sm_lin_quad_cHW
810 prop_binSR_electron2017_bin16_sm_lin_quad_cHW
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
811 prop_binSR_muon2017_bin15_sm_lin_quad_cHW
812 prop_binSR_electron2018_bin16_sm_lin_quad_cHW
813 prop_binSR_electron2017_bin5_sm_lin_quad_cHW
814 prop_binSR_muon2018_bin16_sm_lin_quad_cHW
815 prop_binSR_muon2018_bin10_sm_lin_quad_cHW
816 prop_binSR_muon2016M_bin8_quad_cHW
817 prop_binCRWS_electron2016M_bin16_sm_lin_quad_cHW
818 prop_binCRWS_muon2018_bin15_sm_lin_quad_cHW
819 prop_binSR_muon2016M_bin14_quad_cHW
820 prop_binCRWS_electron2016M_bin8_sm_lin_quad_cHW
821 prop_binSR_muon2017_bin12_quad_cHW
822 prop_binSR_electron2016M_bin16_quad_cHW
823 prop_binSR_muon2017_bin12_sm_lin_quad_cHW
824 prop_binSR_muon2017_bin15_quad_cHW
825 prop_binCRWS_muon2017_bin10_sm
826 prop_binCRWS_muon2018_bin6_sm_lin_quad_cHW
827 prop_binCRWS_muon2017_bin17_quad_cHW
828 prop_binCRWS_muon2017_bin8_sm_lin_quad_cHW
829 prop_binCRWS_muon2017_bin10_sm_lin_quad_cHW
830 prop_binSR_electron2017_bin5_quad_cHW
831 prop_binSR_electron2017_bin20_quad_cHW
832 prop_binCRWS_muon2016M_bin5_sm_lin_quad_cHW
833 prop_binCRWS_electron2016M_bin15_sm_lin_quad_cHW
834 prop_binCRWS_muon2018_bin5_sm_lin_quad_cHW
835 prop_binSR_electron2016M_bin11_quad_cHW
836 prop_binSR_electron2017_bin9_sm_lin_quad_cHW
837 prop_binCRWS_electron2016M_bin17_quad_cHW
838 prop_binSR_electron2017_bin17_quad_cHW
839 prop_binSR_muon2016M_bin10_quad_cHW
840 prop_binCRWS_electron2016M_bin18_sm_lin_quad_cHW
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
k_cHW = 0.0 +3.2 −3.3
841 prop_binCRWS_muon2016M_bin7_quad_cHW
842 prop_binCRWS_electron2017_bin14_sm_lin_quad_cHW
843 prop_binCRWS_muon2017_bin18_quad_cHW
844 prop_binSR_electron2018_bin14_quad_cHW
845 prop_binSR_muon2017_bin11_quad_cHW
846 prop_binCRWS_electron2016M_bin18_quad_cHW
847 prop_binSR_muon2017_bin18_quad_cHW
848 prop_binSR_electron2016M_bin20_quad_cHW
849 prop_binSR_electron2016M_bin18_quad_cHW
850 prop_binCRWS_electron2018_bin12_sm_lin_quad_cHW
851 prop_binSR_electron2017_bin15_quad_cHW
852 prop_binSR_muon2018_bin11_quad_cHW
853 prop_binSR_electron2016M_bin12_sm_lin_quad_cHW
854 prop_binCRWS_electron2016M_bin20_quad_cHW
855 prop_binSR_muon2017_bin9_sm_lin_quad_cHW
856 prop_binSR_muon2016M_bin15_quad_cHW
857 prop_binSR_electron2017_bin13_quad_cHW
858 prop_binSR_electron2017_bin13_sm_lin_quad_cHW
859 prop_binSR_muon2016M_bin11_sm_lin_quad_cHW
860 prop_binSR_electron2018_bin14_sm_lin_quad_cHW
861 prop_binSR_muon2016M_bin10_sm_lin_quad_cHW
862 prop_binSR_muon2016M_bin16_sm_lin_quad_cHW
863 prop_binSR_muon2016M_bin20_sm_lin_quad_cHW
864 prop_binSR_muon2016M_bin14_sm_lin_quad_cHW
865 prop_binSR_electron2017_bin17_sm_lin_quad_cHW
866 prop_binSR_electron2018_bin18_sm_lin_quad_cHW
867 prop_binSR_muon2017_bin13_sm_lin_quad_cHW
868 prop_binCRWS_muon2017_bin17_sm_lin_quad_cHW
869 prop_binSR_electron2016M_bin12_quad_cHW
870 prop_binSR_electron2018_bin18_quad_cHW
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δk_cHW



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\widehat{k\_cHW} = 0.0^{+3.2}_{-3.3}$
871 prop_binSR_electron2017_bin11_sm_lin_quad_cHW
872 prop_binCRWS_muon2017_bin18_sm_lin_quad_cHW
873 prop_binCRWS_muon2016M_bin5_quad_cHW
874 prop_binCRWS_muon2017_bin10_quad_cHW
875 prop_binSR_electron2017_bin10_quad_cHW
876 prop_binSR_muon2017_bin13_quad_cHW
877 prop_binCRWS_electron2018_bin19_sm_lin_quad_cHW
878 prop_binSR_muon2016M_bin20_quad_cHW
879 prop_binCRWS_muon2017_bin18_sm
880 prop_binCRWS_muon2017_bin14_quad_cHW
881 prop_binCRWS_electron2018_bin19_quad_cHW
882 prop_binSR_muon2017_bin9_quad_cHW
883 prop_binSR_muon2016M_bin17_quad_cHW
884 prop_binSR_muon2016M_bin17_sm_lin_quad_cHW
885 prop_binSR_electron2016M_bin20_sm_lin_quad_cHW
886 prop_binSR_muon2016M_bin13_quad_cHW
887 prop_binSR_electron2016M_bin18_sm_lin_quad_cHW
888 prop_binSR_muon2016M_bin13_sm_lin_quad_cHW
889 prop_binCRWS_muon2017_bin14_sm_lin_quad_cHW
890 prop_binSR_electron2017_bin18_quad_cHW
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\widehat{k\_cHW}$